The Big Pool Bangs Theory

Getting Rid of Dark Matter

Abstract. One of the two dark sectors can be dispensed with, the other cannot. In the framework presented here dark matter needs no new particle: it is the $88\% \pm 4\%$ of the original Planck-mass relic mass that a chaotic post-bounce rebound throws outward, ordinary gravitating matter invisible to every non-gravitational detector, so the forty-year null of direct searches becomes a prediction (P5 (No Particle DM)) and a confirmed particle detection would falsify the core hypothesis. Dark energy is *not* dispensed with: no arrangement of ejecta or of neighbouring universes can accelerate the expansion, and the acceleration is attributed to a constant tension of space whose value the framework does not derive. With that scorecard stated, we explore a mechanical alternative to inflationary ΛCDM in which the universe begins as a jammed ensemble of Planck-mass relics ($\sim 10^{60}$–10^{61} per present Hubble volume), treated as effective hard spheres (an ansatz motivated by the Compton–Schwarzschild coincidence).[1] A loop-quantum-cosmology-type modified Friedmann equation, with critical density conjecturally identified with the hard-sphere jamming density, produces a non-singular bounce; the subsequent chaotic rebound ejects most of the ensemble. Toy N-body simulations with a turbulent kick distribution eject $88\% \pm 4\%$ of the mass, consistent with the observed dark-matter fraction for $\sigma_{\rm rel} \approx 1$ — a regime a cascade nucleation model only marginally reaches — and show clean velocity sorting of the ejecta. Two geometric readings of that sorting are carried in parallel: Geometry A (homogeneous), in which the jammed patch is super-horizon and the sorting is velocity-space structure whose initial peculiar velocities must redshift into the Hubble flow; and Geometry B (finite centred cloud), in which the rebound is a single centred event and the radial ordering survives as a ballistic Hubble-like law. Geometry B currently has *no bounce mechanism* — a finite cloud of the relevant mass lies $\sim 10^{40}$ inside its own Schwarzschild radius and the homogeneous bounce equation offers it no exit — and is retained only as the geometry that our-position tests can probe kinematically; its monopole distance–redshift relation with the tension included has not been computed. The framework does *not* reproduce cosmic acceleration: coasting expansion is disfavored by Pantheon+ ($\Delta\chi^2 \approx 84$) and DES-SN5YR ($\Delta\chi^2 \approx 131$), a relativistic LTB lightcone through the ejecta decelerates ($q_0^{\rm eff} = +0.16$), a toy effective $w(z)$ drifts in the wrong direction relative to the DES+DESI fit, and a kinetic-Sunyaev–Zel'dovich gate shows that no ejection profile can supply more than $\sim 1\%$ of the observed dark energy. We therefore adopt the Big Pool Bang framework *plus a constant tension of space* ($w = -1$) as the expansion history: the rebound sets the origin and initial velocity structure of the matter, the tension sets the acceleration, and the value of the tension is not derived. A predicted Cold Spot hot ring at 5ř–7ř is absent in Planck 2018 data. Because a neighbouring universe imprinting the last-scattering surface lies beyond our particle horizon, only an initial-condition (Sachs–Wolfe) imprint of its potential is causally admissible; for the Cold Spot this fixes a source mass $M_2 \approx 10^{48}$ kg ($+0.6/-0.4$ dex, an upper bound at fixed

[1] Code, figures, build instructions and download links for all datasets used: https://github.com/AIMFIRST-VN/big001

amplitude) 1–2.5 Gly beyond last scattering, whose induced velocity of the local volume is $\lesssim 1$ km s^{-1}. We organize the claims into seven hypotheses[2]: the bounce-and-ejection core, H1 (Bounce); the Cold Spot as a frozen vortex, H2 (Vortex); a neighbouring Big Pool Bang in the Cold Spot direction, H3 (Neighbour); a population of such neighbours, H4 (Population); our own position inside Universe 1, H5 (Our Position), which is the discriminator between Geometries A and B — a structure-subtracted off-centre dipole above the 95th-percentile mock floor (≈ 2.5 Mpc at present) or a bulk flow growing with depth supports B, an offset consistent with the mock distribution together with a decaying flow favours A; the location of the second Big Pool Bang, H6 (Neighbour Location); and the population model, H7 (Pool Model). Each carries explicit falsifiers collected in one table; we catalogue ten predictions (P1 (Polarization Ring)–P10 (Induced-Velocity Bound)) with the instruments required, and state the unsolved problems: causal structure of the bounce, relativistic formulation of ejection, the relativistic (hot) temperature of the ejecta, the horizon and flatness problems, and the expansion history.

1. Introduction: Three Questions

This paper is organized around three questions.

1. **Where are we?** If the universe began as a jammed ensemble of Planck-mass relics that rebounded and ejected most of its mass, the ejection was velocity-sorted (Sec. 3, Fig. 2). What that sorting means depends on the geometry, and this paper carries two readings (Sec. 3). In *Geometry A* (homogeneous) the jammed patch is super-horizon, ejection happens everywhere, and the sorting is velocity-space structure on an FLRW background: the ejection sets the matter's initial peculiar-velocity distribution, which redshifts into the Hubble flow only if the cooling problem of Sec. 12 is solved. In *Geometry B* (finite centred cloud) the rebound is a single centred event, genuine radial ordering survives — the fastest furthest — and the Hubble-like law is ballistic ($v = r/t$) plus the adopted tension — though Geometry B has, at present, no bounce mechanism (Sec. 3). H5 (Our Position; Sec. 10) is the discriminator: a displaced-centre dipole above the mock floor or a bulk flow that grows with depth supports B; an offset consistent with the mock distribution together with a decaying flow favours A. Our current answer: the displaced-centre test is null to within a few Mpc, the bulk flow at ~ 430 km s^{-1} is the only kinematic handle, and the hemispherical asymmetry is better assigned to an external neighbour than to our own offset.

2. **Where is the second Big Bang?** A neighbouring Big Pool Bang beyond our particle horizon can imprint the last-scattering surface only through the potential it contributed to the initial conditions — a Sachs–Wolfe spot with a low-multipole tail. The Cold Spot at (l, b) = (203ř, −56ř) is the candidate, treated under H2 (Vortex), H3 (Neighbour) and H6 (Neighbour Location) in Secs. 6–8: direction known to ±3ř, comoving distance 1–2.5 Gly beyond the last-scattering surface, mass $M_2 \approx 10^{48}$ kg to within $+0.6/-0.4$ dex, and a curl-mode test, P8 (Curl Test), pre-registered. The eight extreme spots of the Planck map are the candidate list for a population of such neighbours, H4 (Population) and H7 (Pool Model), treated in Sec. 9.

3. **How does the Big Pool Bang model work, and what does it explain?** A first-order vacuum crystallization jams into a hard-sphere state; a loop-quantum-cosmology-type bounce rebounds it; chaotic ejection sends $88\% \pm 4\%$ of the mass outward as dark matter; this is H1

[2]The author's working nicknames for five of the hypotheses, retained here for continuity with earlier drafts: H1 (Bounce) "The Big Pool Bang"; H2 (Vortex) "The Cold Eye of Sauron"; H3 (Neighbour) "Big Pool Bang Multiverse 2"; H4 (Population) "The Dark Comic Background"; H7 (Pool Model) "The Pool Model". Headings use plain titles.

(Bounce), Secs. 2–4. In Geometry A the ejection fixes the matter's initial peculiar velocities on an assumed background; in Geometry B it is a centred event whose radial ordering is the Hubble-like law itself. The rebound kinetics do *not* reproduce cosmic acceleration in either geometry, so the adopted expansion history is the Big Pool Bang framework plus a constant tension of space whose value is not derived (Sec. 5.2); the predicted Cold Spot thermal ring is absent, and the ejecta are relativistic, so "cold" dark matter is unproven (Sec. 12).

The paper answers the third question first, then treats the Cold Spot (Sec. 6), the neighbour mechanism (Sec. 7), its location (Sec. 8), the population (Sec. 9), and our own position (Sec. 10). Section 11 collects the ten predictions and the single falsification table; Sec. 12 lists open problems.

1.1 Vacuum Crystallization and the Hard-Sphere State

Standard cosmology relies on scalar-field inflation to explain why the observable universe is flat, isotropic, and thermally uniform, at the cost of fine-tuning and of singularity theorems (Borde, Guth & Vilenkin 2003). The **Big Pool Bangs Theory** is a mechanical alternative grounded in first-order phase-transition dynamics. Space begins in a false vacuum with $\rho_{\text{vac}} \sim \rho_P$. Quantum fluctuations trigger a **vacuum crystallization** transition which nucleates at many points: bubbles of converted phase grow, collide, and merge, and the "bang" is their percolation into a single jammed state.

Timeline. Bubble walls propagate at $\leq c$. The comoving region that becomes one present-day Hubble volume spans, at the bounce density, $\sim 10^{20}$ causal (Planck-size) patches per dimension, so percolating it takes at least $10^{20}\, t_P = 5.4 \times 10^{-24}$ s divided by the number of nucleation sites per crossing length; any nucleation density sparse enough to give nontrivial merger statistics implies an assembly time many orders above t_P, whose consistency with trapped-surface avoidance (Sec. 3) is open.

Nucleation as the origin of the kick spectrum. The randomness of the merger history is inherited by the jammed core as internal turbulence, a candidate source for the stochastic kick spectrum that Sec. 3 otherwise requires as an input — provided nucleation-era velocity structure survives the collisional jammed epoch rather than thermalizing. A Poisson nucleation-and-merger simulation (vector-summed wall impulses) recovers a Maxwellian kick distribution (speed coefficient of variation 0.41–0.42), largely a central-limit consequence; the nontrivial statistic is the coherent-to-dispersive ratio σ_{rel}. Poisson nucleation is turbulence-dominated ($\sigma/\langle v_r \rangle \approx 5$–$8$) and cannot supply the coherent radial component the ejection fraction requires. A *cascade* model — a single seed bubble whose wall catalyzes further nucleation, halted by a jamming feedback at $\eta \approx 0.64$ — yields $\sigma_{\text{rel}} = 0.65$–$0.78$, robust across a $4\times$ range in trigger distance, against the ≈ 0.75–1.0 window the dark-matter fraction requires: the cascade reaches that window only marginally. Three caveats gate the chain: the $88\% \pm 4\%$ figure comes from Maxwellian-ansatz N-body runs and the cascade's heavier-tailed kick field has not been fed through the N-body pipeline; the model retains structural choices (trigger distance, nucleation lag, wall-area weighting, the arrest $\eta = 0.64$); and the wall-momentum efficiency is assumed universal, whereas in first-order-transition physics it is model-dependent and typically small (Espinosa et al. 2010). Bubble collisions in any first-order transition also source a stochastic gravitational-wave background peaked at the transition scale; this is a generic prediction of H1 (Bounce), listed as P3 (Transition GW Background), not of any neighbour.

Because quantum geometry prohibits localized infinite densities, the energy crystallizes into irreducible **Planck-mass relics** ($m_P \approx 21.8\,\mu$g). We count them *per present-day Hubble volume*: $N \approx 10^{60}$–10^{61} relics, of mass $M_1 = N m_P \approx 2.2 \times 10^{52}$ kg at the lower end (order of magnitude; the matter mass of the *observable* universe is $\sim 1.5 \times 10^{53}$ kg). These are the relic number and mass per Hubble volume today, not totals: in Geometry A (Sec. 3) the jammed patch is super-horizon, so its total number and mass are unbounded by observation; in Geometry B they are the mass of a finite cloud, at least the $\sim 1.5 \times 10^{53}$ kg of the observable universe. The relics saturate local volume up to the random close-packing limit $\eta_{\rm rcp} \approx 0.64$, and with $\rho_P = m_P/\ell_P^3 = 5.2 \times 10^{96}$ kg m^{-3} the jamming density is $\rho_c = \eta_{\rm rcp}\rho_P$. Compressed to ρ_c, one Hubble volume's worth of relics occupies a region $\sim 10^{-15}$ m $\approx 10^{20}\,\ell_P$ across — but this must not be read as the size of an isolated object: the Schwarzschild radius of 2.2×10^{52} kg is 3.3×10^{25} m, forty orders larger, so a finite ball of this mass and density would be a black hole, not a bouncing patch. The bounce of Sec. 3 is meaningful only if the jammed patch is the whole (or a super-horizon) universe described by a homogeneous FLRW-type metric, for which no exterior Schwarzschild geometry exists (Geometry A of Sec. 3); the finite-cloud reading (Geometry B) must confront this trapped-surface problem directly. We therefore use ρ_c, not a jamming radius, as the physical parameter.

2. Planck-Mass Relics as Effective Hard Spheres

For a particle of mass m the reduced Compton wavelength $\lambda_C = \hbar/mc$ and the Schwarzschild radius $R_s = 2Gm/c^2$ coincide at $m \approx m_P$, where both equal $\sim \ell_P = 1.616 \times 10^{-35}$ m. This is the regime in which neither quantum field theory nor classical general relativity is separately valid. **We adopt, as a modeling ansatz rather than a derivation, that Planck relics possess a minimum spatial extent $\sim \ell_P$ that no interaction can compress**, and treat them as effective incompressible hard spheres. The ansatz is judged by the phenomenology it generates (Secs. 3–10); its microphysical justification is an open problem (Sec. 12).

3. H1 (Bounce): The Kinetic Bounce and Chaotic Ejection

Why hard-sphere pressure cannot bounce the collapse. Inside a homogeneous fluid the dynamics are governed by the Raychaudhuri equation

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3P}{c^2}\right).$$

Near random close packing the hard-sphere kinetic pressure diverges (free-volume equation of state, $P \propto nk_BT\,[1 - (\eta/\eta_{\rm rcp})^{1/3}]^{-1}$), but in general relativity positive pressure *gravitates*: no classical equation of state with $P > 0$ can bounce an FLRW collapse. The rebound must be sourced by quantum-geometric corrections that dominate at the jamming density $\rho_c = \eta_{\rm rcp}\,\rho_P$. The minimal object with the required property is the loop-quantum-cosmology (LQC) modified Friedmann equation,

$$H^2 = \frac{8\pi G}{3}\,\rho\left(1 - \frac{\rho}{\rho_c}\right), \qquad \rho_c = \eta_{\rm rcp}\,\rho_P,$$

which encodes a maximum attainable density covariantly: $H \to 0$ as $\rho \to \rho_c$ and the effective Raychaudhuri equation flips sign. We propose the jamming limit of the relic gas as the physical origin of the ρ_c that LQC introduces axiomatically, noting that LQC's own Immirzi-fixed value is $\rho_c \approx 0.41\,\rho_P$ versus $0.64\,\rho_P$ here — order-of-magnitude agreement, a coincidence rather than a derivation. A numerical integration (Fig. 1) confirms that the classical hard-sphere fluid overshoots the jamming density by $> 10^4$ while the modified equation bounces at $\rho = \rho_c$; the bounce is built into the equation, not independently confirmed by it. Whether it completes before a trapped surface forms around the still-homogeneous patch is an assumption (Sec. 12).

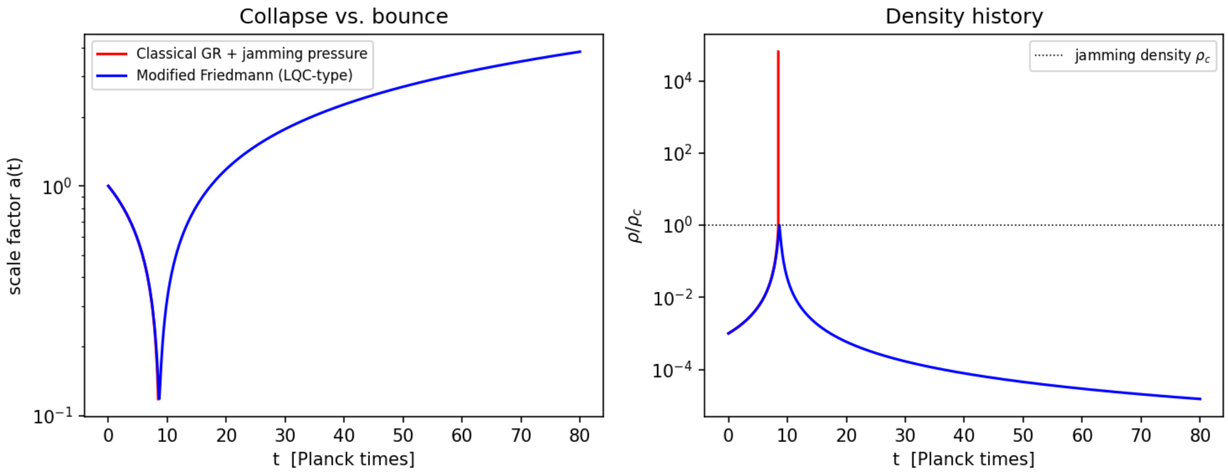

Figure 1: Fig. 1 — Toy FLRW integration. Left: scale factor. Right: density history. Classical GR with jamming pressure (red) overshoots the jamming density by $> 10^4$ (collapse); the modified Friedmann equation (blue) bounces at ρ_c.

Regime disclaimer. Everything that follows in this section is a **Newtonian analogy**: the real dynamics occur where escape velocities formally exceed c by $\sim$20 orders of magnitude and only a general-relativistic treatment is meaningful. We use the Newtonian model because violent-relaxation statistics (which fraction unbinds, how ejecta sort by speed) are plausibly governed by dimensionless energy ratios that survive the translation; whether they do is the central open problem of Sec. 12.

The slingshot phase. Picture the jammed core as 10^{60} billiard balls packed to the jamming limit; when the bounce reverses the collapse, the rebound plays the role of the opening break. At jamming density the relic gas is collisional, so the slingshot phase begins only after expansion dilutes the ensemble below the contact threshold and relics decouple onto ballistic trajectories. From then on the system is a dense self-gravitating N-body ensemble in which three-body encounters dominate relaxation — the "core collapse and halo ejection" of globular-cluster dynamics, which sheds most of a system's mass to reach a bound remnant. Applied to the birth of the universe it sorts the mass into two components:

- **The ejected halo ($\approx$ 88%):** relics kicked above escape velocity, free-streaming and rarely colliding again — the candidate dark-matter component, requiring no new particle (but see the temperature caveat below).
- **The trapped furnace ($\approx$ 12%):** relics that fall back and grind against each other in a turbulent core, proposed as the origin of the photon–baryon plasma. **Caveat (essential):** this requires an unspecified conversion channel from Planck relics to Standard-Model plasma — including baryogenesis and the photon-to-baryon ratio — while ejected relics remain stable for 13.8 Gyr. Absent such a channel (e.g. a threshold collision energy the ejecta never reach), the trapped fraction cannot be identified with the baryon budget.

Ejecta temperature. The ejecta leave at a substantial fraction of the (formally superluminal) escape speed, i.e. relativistically. Without a redshifting or cooling argument that brings their thermal velocities below the free-streaming limit by the epoch of structure formation, Planck-relic ejecta behave as *hot*, not cold, dark matter. Cosmological redshifting of free-streaming momenta ($p \propto 1/a$)

is the natural candidate, but no calculation of the residual dispersion at matter–radiation equality exists here; this is a first-order open problem (Sec. 12).

Ejection fraction (toy N-body). A softened $N = 1000$ equal-mass simulation of a cold uniform sphere given a radial kick $v = f\, v_{\rm esc}$ (with $v_{\rm esc}$ the surface escape speed) has a steep ejection threshold: $f = 0.6 \to 3\%$, $0.8 \to 41\%$, $0.9 \to 98\%$, $f \geq 1 \to 100\%$. A monolithic kick reproduces the dark-matter fraction only in the narrow window $f \approx 0.85$–0.9, and the surface-escape-speed kick ($f = 1$) over-ejects. A Maxwellian turbulent component of dispersion $\sigma_{\rm rel}\bar{f} v_{\rm esc}$ superimposed on a mean rebound $\bar{f} v_{\rm esc}$ broadens the transition: at $\bar{f} = 0.7$ the ejected fraction rises from 22% ($\sigma_{\rm rel} = 0.5$) through $\approx 88\%$ (1.0) to 98% (1.5). Over a 15-seed ensemble at $(\bar{f}, \sigma_{\rm rel}) = (0.7, 1.0)$ the ejected fraction is $88\% \pm 4\%$, consistent with the observed dark-matter mass fraction, at a $\sigma_{\rm rel}$ the cascade model of Sec. 1 only marginally reaches; $(\bar{f}, \sigma_{\rm rel})$ remain inputs to be derived from the bounce turbulence spectrum.

Ejecta kinematic structure (15 seeds, 13{,}140 ejecta). Ejection is extended (median ejection time ~ 1 dynamical time; 10% of ejecta leave after 11), with clean monotonic velocity sorting: median speeds fall from $0.71\, v_{\rm esc}$ for the earliest ejecta to $0.28\, v_{\rm esc}$ for the latest (Fig. 2), so radial position maps onto ejection epoch. The tangential structure shows no correlated anisotropy: the per-sky ejection dipole (0.064 ± 0.024) matches shot noise (0.053 ± 0.024), and early-vs-late shell axes align only weakly ($\langle \cos\theta \rangle = 0.37 \pm 0.49$). The isotropic cascade therefore supplies no preferred axis; if one exists it must be seeded externally (Sec. 7).

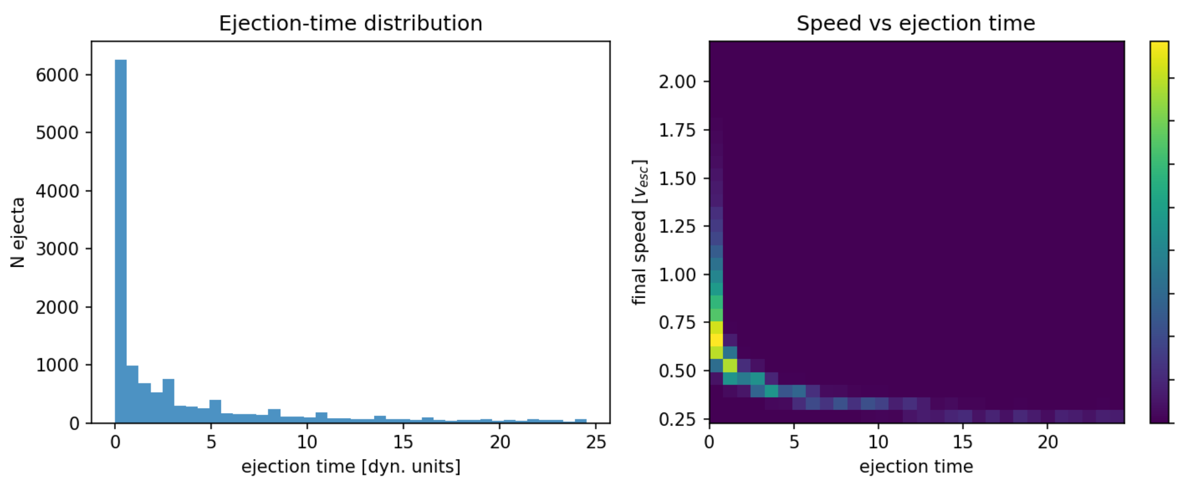

Figure 2: Fig. 2 — Ejecta kinematics over 15 seeds: ejection-time distribution (left) and speed vs ejection time (right), showing the velocity-sorted shell structure.

From the N-body cloud to the universe: two geometries. The N-body ejecta cloud is a sub-horizon toy of the ejection *mechanism*. How it maps onto the universe is not settled by the toy, and this paper carries two geometric hypotheses in parallel.

- **Geometry A (homogeneous).** The jammed patch is the whole (super-horizon) universe (Sec. 1), there is no exterior, and ejection happens everywhere: every comoving region is simultaneously "furnace" and "halo". The radial shells of Fig. 2 translate into *local velocity-space structure* on an assumed FLRW background, not into a centred cloud with an edge. What survives is the statistical content (ejected fraction, speed distribution, isotropy); what does not is the geometric centre. The ejection sets the matter's *initial peculiar-velocity distribution*, which redshifts into the Hubble flow only if the cooling problem of Sec. 12 is solved. This geometry avoids the trapped-surface problem of Sec. 1.1 and predicts a null for H5 (Our Position).

- **Geometry B (finite centred cloud).** The rebound is a single centred event and genuine radial velocity sorting survives: the fastest furthest. The Hubble-like law is ballistic, $v = r/t$, plus the adopted tension of Sec. 5.2. Two burdens must be stated plainly. First, **Geometry B currently has no bounce mechanism.** A finite cloud of the relevant mass ($M_1 = 2.2 \times 10^{52}$ kg per Hubble volume, and at least the 1.5×10^{53} kg of the observable universe) compressed to ρ_c is $\sim 10^{-15}$ m across against a Schwarzschild radius of $\sim 10^{25}$ m — some 10^{40} inside its own horizon — and the homogeneous LQC equation above, written for a metric with no exterior, offers such a cloud no exit: it is a trapped region, and nothing in this paper takes it out again. Geometry B is retained not because a bounce has been shown to occur in it but because it is the geometry that H5 (Our Position) can test kinematically. Second, its principal observational burden is the *monopole* distance–redshift relation, not the dipole: a few-Mpc offset in a Gpc-scale cloud is trivially isotropic, so supernova isotropy is not the issue; the issue is that the ballistic-only LTB lightcone through the ejecta (Appendix C) decelerates ($q_0^{\text{eff}} = +0.16$, ~ 0.35 mag from ΛCDM by $z \sim 5$), and the LTB-plus-tension lightcone that Geometry B actually requires has not been computed (Sec. 5.2). What H5 (Our Position) directly tests is our offset from the centre and radial ordering in the bulk flow.

H5 (Sec. 10) is the discriminator between A and B: a structure-subtracted off-centre dipole above the 95th-percentile mock floor with a direction stable across shells, or a bulk flow that grows with depth, supports B; an offset consistent with the mock distribution together with a flow that decays with depth favours A.

4. Consistency Checks

- **Horizon problem — open.** At the bounce the causal patch is of Planck size, and the region that becomes one present-day Hubble volume comprises $\sim 10^{60}$ such patches (of order one per relic). Nothing in the bounce equilibrates them within a few Planck times, so homogeneity across causal patches is an assumed initial condition — in Geometry A over a super-horizon, unbounded patch; in Geometry B over the finite cloud. A long contracting phase before the bounce could in principle enlarge the causal patch, but the collapse is not modelled here.
- **Flatness — not addressed.** In the turbulent regime the simulations favor ($\bar{f} = 0.7$, $\sigma_{\text{rel}} = 1.0$) the mean kinetic energy is $\langle v^2 \rangle = \bar{f}^2(1 + \sigma_{\text{rel}}^2) v_{\text{esc}}^2 \approx v_{\text{esc}}^2$, but this does *not* make the total energy vanish. For a uniform sphere $U = -3GM^2/5R$ while the surface escape speed gives $v_{\text{esc}}^2 = 2GM/R$, so $\langle v^2 \rangle = v_{\text{esc}}^2$ means $K \approx 1.7\,|U|$ (or $K \approx 2\,|U|$ if the kick is scaled to the local escape speed); $E = 0$ would require $\bar{f}^2(1 + \sigma_{\text{rel}}^2) \approx 0.5$–$0.6$ (of order 0.3–0.6 depending on the escape-speed convention and kick weighting). The simulated systems are therefore unbound, $E > 0$, which is consistent with the 88% ejected fraction and with the surface-escape-speed kick $f = 1$ ejecting 100%. The energy budget of the rebound says nothing about spatial flatness; flatness is not addressed by this framework.

Remark on the spectral tilt. One may parametrize the scalar index by a turbulent dissipation ratio

ν_t (eddy turnover time to expansion time) as $n_s - 1 = -\nu_t/3$; Planck's $n_s = 0.965 \pm 0.004$ then gives $\nu_t \approx 0.105$. Both the relation and the value are fitted inputs, and showing that a Kolmogorov cascade ($E(k) \propto k^{-5/3}$) can produce a nearly scale-invariant curvature spectrum is an open problem (Sec. 12).

5. Expansion History: Why the Big Pool Bang Framework Needs a Constant Tension

The rebound kinetics alone imply a coasting expansion ($a \propto t$), and this is excluded. Against 1371 Hubble-flow SNe Ia from Pantheon+ (Scolnic et al. 2022; Brout et al. 2022; diagonal errors, offset marginalized), coasting yields $\chi^2 = 674$ vs 590 for flat ΛCDM ($\Delta\chi^2 \approx 84$), though far better than Einstein–de Sitter ($\chi^2 = 1170$; Fig. 3); on DES-SN5YR (DES Collaboration 2024; Dovekie recalibration, Popovic et al. 2025; 1820 SNe, full covariance) the figures are 1632, 1763 ($\Delta\chi^2 \approx 131$), and 2566. Whether the *distribution* of the ejecta could mimic acceleration for an interior observer is tested in Appendix C: a relativistic Lemaître–Tolman–Bondi lightcone through the measured ejecta shells decelerates ($q_0^{\mathrm{eff}} = +0.16$); no smooth, bimodal, or lumpy ejection profile accelerates; a blast-wave cavity is the giant-void geometry; and the kinetic Sunyaev–Zel'dovich limit on coherent cluster flows ($v \lesssim 300$ km s^{-1} at Gpc distances) caps any cavity's contribution at $|\Delta q_0| \lesssim 0.01$ against the $\Delta q_0 \approx -1.05$ required. **No ejection profile of any morphology supplies more than $\sim$1% of the observed dark energy.**

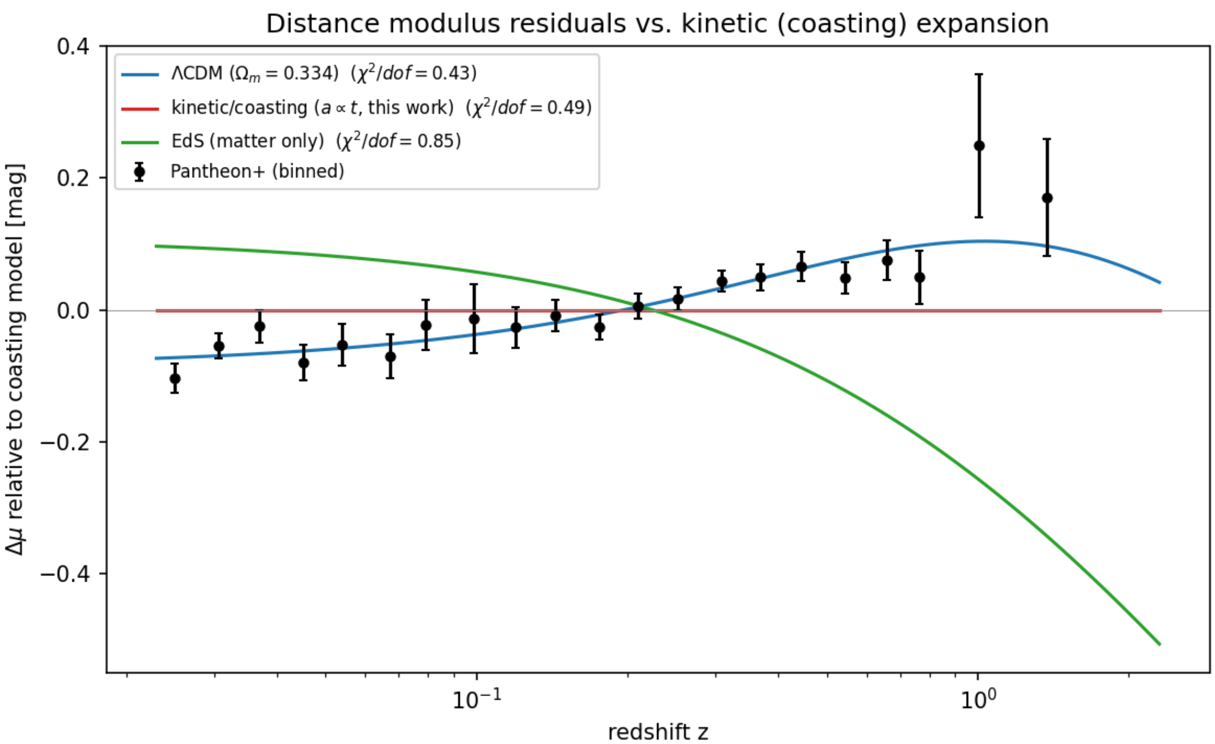

Figure 3: Fig. 3 — Pantheon+ binned distance-modulus residuals relative to the coasting model.

5.1 The $w(z)$ drift

Integrating the measured ejecta speed distribution as cold spherical shells decelerating in their enclosed mass, and computing the equation of state an interior observer would fit, gives $w_{\rm eff}$ drifting from -0.29 at $z = 2$ to -0.31 today (Fig. 4): a few-percent drift near the coasting value $-1/3$. Because this apparent equation of state is set by the evolving ratio of velocity-dispersion energy to clustered rest-mass energy, it cannot be constant. The DES-SN5YR + DESI DR2 BAO (+CMB) w_0w_aCDM fit prefers evolving w at $\sim 3\sigma$ ($w_0 \approx -0.8$, $w_a \approx -0.7$; DES Collaboration 2024; DESI Collaboration 2025), with w becoming *less* negative with time; the toy shells drift the *opposite* way. The remap is crude (91% of the ejecta re-bind in a static enclosed-mass model, so the flow is built from the fastest 9%), but as it stands the kinetic sector shares with the data only the qualitative statement that w evolves, and predicts the wrong sign of the drift.

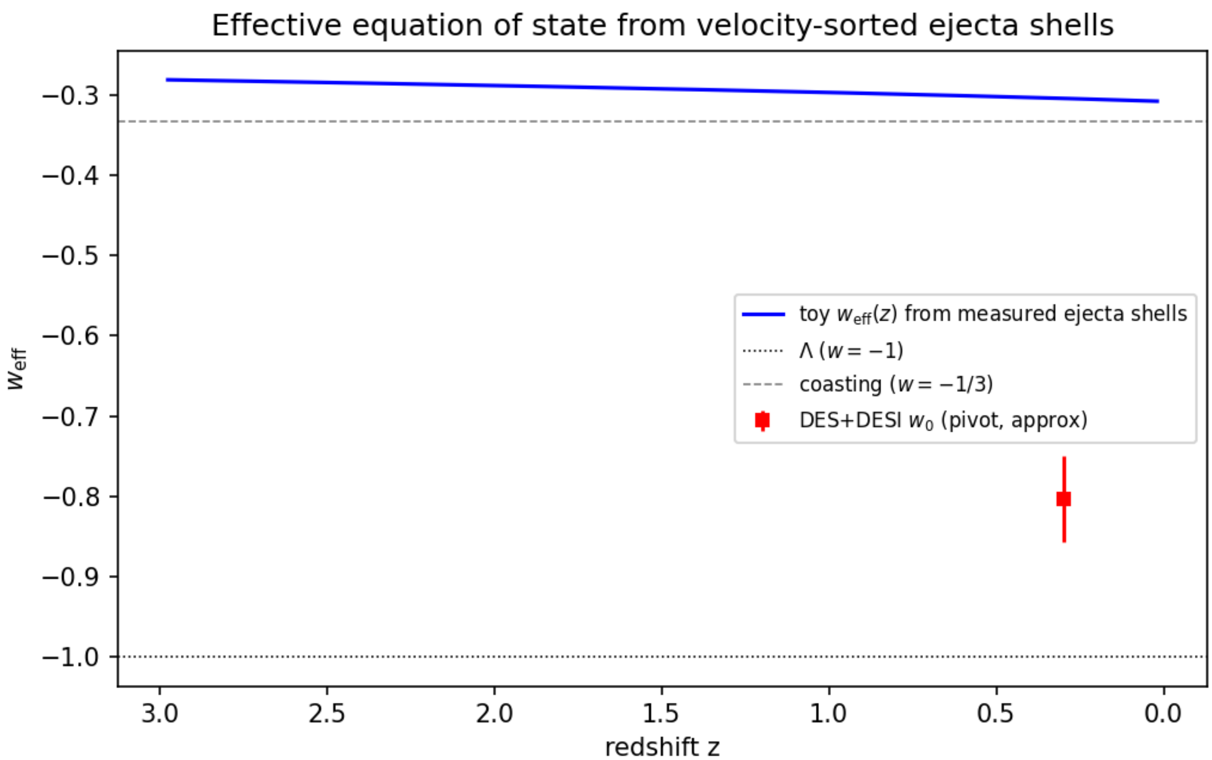

Figure 4: Fig. 4 — Toy effective equation of state from velocity-sorted ejecta shells: a few-percent drift near $w = -1/3$, in the opposite direction to the DES+DESI w_0w_a best fit, and of insufficient magnitude.

5.2 The adopted expansion history: the Big Pool Bang framework plus a constant tension

The failures above are failures of one idea only: that the *kinetics* of the rebound could substitute for dark energy. They do not touch the bounce, the ejection, or the dark-matter fraction. We therefore separate the two sectors. The Big Pool Bang framework is responsible for the origin and velocity structure of the matter; the acceleration is attributed to a constant tension of space, $\rho_\Lambda c^2 = -p_\Lambda$, which we adopt without claiming to derive it.

Why a tension. For a comoving separation $d = a\chi$, $\ddot{d} = (\ddot{a}/a)\,d$ with $\ddot{a}/a = -\frac{4\pi G}{3}(\rho + 3p/c^2)$: acceleration is proportional to distance, with a coefficient set locally by density and pressure, and is repulsive only for $\rho + 3p/c^2 < 0$. If space stores an energy density $u(a)$ when stretched, thermodynamics ($p = -\partial(uV)/\partial V$) fixes its pressure: for $u \propto a^{-n}$, $w = n/3 - 1$. Matter ($n = 3$) attracts; strings or curvature ($n = 2$, $w = -1/3$) coast; a domain-wall *network* ($n = 1$, $w = -2/3$) repels; a constant tension ($n = 0$, $w = -1$) repels; an energy growing as a^2 ($w = -5/3$) is phantom. The wall-network value applies to many walls per Hubble volume (Bucher & Spergel 1999); a single wall at $\sim$46 Gly contributes no such term. Observationally, Planck+BAO+SN constrain a constant equation of state to $w = -1.03 \pm 0.03$ (Planck Collaboration 2020 VI), while the DES+DESI $w_0 w_a$ fit prefers evolving w at $\sim 3\sigma$ (Sec. 5.1). We adopt constant $w = -1$ as the minimal assumption consistent with the first, noting that the evolving-w hint, if confirmed, is not something the kinetic sector can explain: its toy drift has the wrong sign. Neighbouring Big Pool Bangs cannot contribute: a single neighbour supplies a tidal shear and a negligible induced velocity (Sec. 8), a symmetric population supplies nothing by the shell theorem (Appendix B), and neither is a monopole. Today the tension term gives $\ddot{a}/a = +3.4 \times 10^{-36}\ \mathrm{s}^{-2}$ against $-0.7 \times 10^{-36}\ \mathrm{s}^{-2}$ from matter.

Vacuum energies of neighbours do not sum. Vacuum energy acts only where it resides: the expansion rate of our region is fixed by the tension inside it, so the tensions of any number of neighbouring bangs contribute nothing here, for the same locality reason that their masses contribute no monopole (Appendix B). If all bangs crystallized into the same vacuum, the tension is one universal constant whose value the model does not predict; if into different vacua, the differences reside in the walls between them, and a higher-tension neighbour drives its wall outward as a bubble collision.

The background is assumed, not derived. With the tension included, the ejecta are treated as pressureless dust on a flat ΛCDM background once virialized into halos; the $w_{\rm eff} \approx -0.3$ of the shell toy (Sec. 5.1) describes the shell *kinematics*, not the background equation of state. In Geometry A (Sec. 3) the ejection sets the matter's initial peculiar-velocity distribution on that background, those peculiar velocities redshift into the Hubble flow only if the cooling problem of Sec. 12 is solved, and the supernova, BAO and kSZ comparisons are satisfied because the ΛCDM background is assumed: the ejecta shells of Sec. 3 describe how the matter component acquired its initial velocities rather than an alternative expansion law. Geometry B has no such shelter. There the expansion law *is* the ballistic ordering of the ejecta, $v = r/t$, plus the tension, and the observational comparison must be made on that law. The ballistic-only LTB lightcone (Appendix C) decelerates ($q_0^{\rm eff} = +0.16$) and departs from ΛCDM by $\sim$0.35 mag by $z \sim 5$; the LTB-plus-Λ lightcone — an inhomogeneous ballistic dust cloud with a constant tension, seen by an interior observer — has *not* been computed. That monopole distance–redshift relation, not the dipole, is Geometry B's principal observational burden, and until it is computed Geometry B cannot claim the supernova, BAO and kSZ agreement that Geometry A inherits from its assumed background. What the framework does not supply is the value of the tension: the one framework-native candidate, a residual unconverted vacuum fraction, reproduces the 10^{-123} tuning of standard cosmology (Appendix C), and the origin of ρ_Λ is counted among the open problems (Sec. 12), exactly as in ΛCDM.

6. H2 (Vortex): The Cold Spot and the Axis of Evil

The Eridanus Cold Spot (Vielva et al. 2004; Cruz et al. 2005) is the sky's outstanding non-Gaussian feature. On the Planck SMICA map (Planck Collaboration 2020 I, IV, VII) it is centred at galactic (l, b) = (203ř, −56ř), with a decrement of $-70\,\mu$K at 5ř smoothing and a peak of $-168\,\mu$K at 2ř smoothing, and a half-maximum radius of 2.5ř; we use these values throughout.

Cyclonic vortex reading. Internal quantum turbulence in the bounce creates macro-vortices. In the **Jovian/cyclonic model** these act as thermal dams that choke local heat transport, deform under shear into elongated features aligned along a preferred rotation axis (the "Axis of Evil"), and — without solid-surface friction to dissipate angular momentum — scale up with expansion and freeze into the CMB at decoupling. The Cold Spot is read as such a frozen cyclone.

Failed prediction: the thermal ring. A cyclone diverting thermal energy around its core would produce a concentric hot ring of +20–50 μK at 5ř–7ř. The azimuthally averaged Planck 2018 SMICA profile ($N_{\text{side}} = 128$) around the Cold Spot, with uncertainties from 500 random high-latitude positions (Fig. 5), shows a cold core extending to $\sim$9ř (-3.4σ at 3ř) and **no hot ring at** 5ř–7ř; a +37–52 μK excess at 13ř–15ř is $\sim 2.5\sigma$ before a trials factor of $\sim$10 radial bins. The thermal-ring prediction, the first half of P1 (Polarization Ring), failed. The polarization part survives: a frozen vortex boundary should carry a radial E-mode ring of 1–3 μK at $\sim$5ř–15ř. Planck's Q_r profile is consistent with zero in every 2ř ring to 20ř with per-ring errors of 0.6–2.7 μK (Appendix A), so the ring is neither detected nor excluded; the test requires CMB-S4 sensitivity.

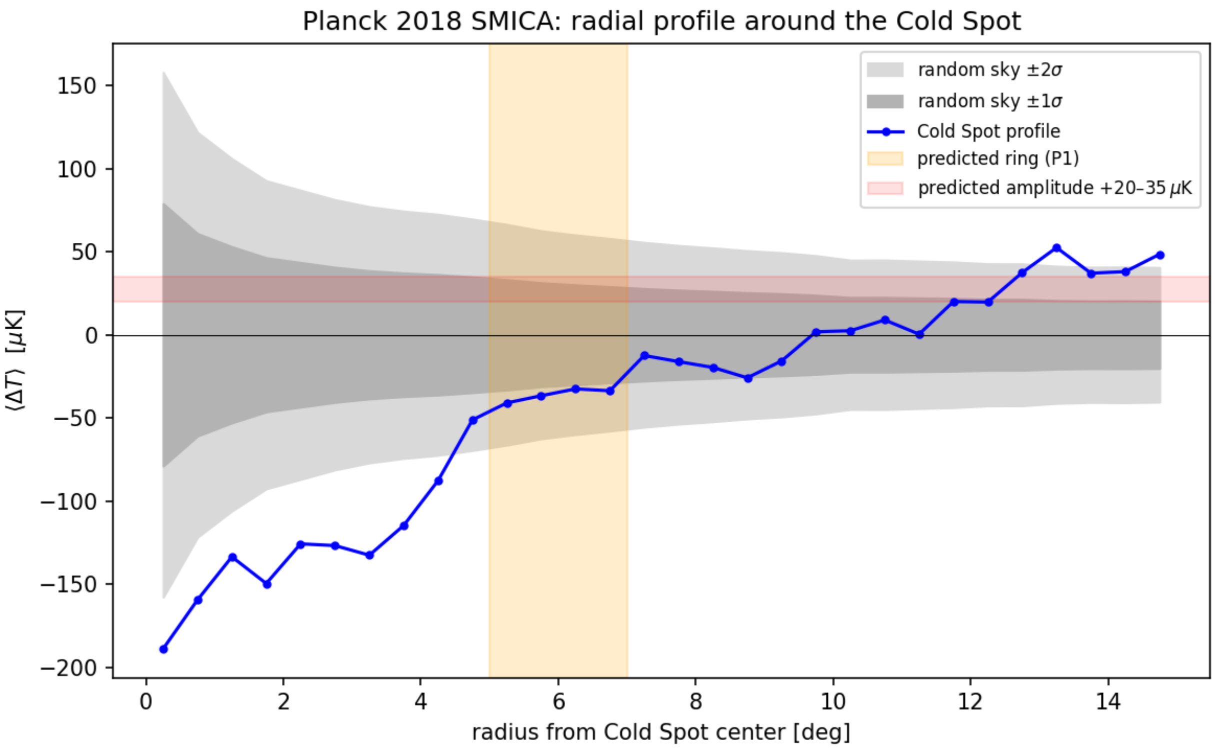

Figure 5: Fig. 5 — Planck 2018 SMICA radial temperature profile around the Cold Spot, vs. 500 random reference positions.

Preferred axis. Our simulations find ejection isotropic within shot noise (Sec. 3), so an axis can only be supplied by an external seed, the neighbouring bang of Sec. 7. The literature's cluster of tentative ($\lesssim 3\sigma$) alignments — a supernova acceleration dipole roughly along our CMB motion (Colin et al. 2019; Sah & Rameez 2024), the quadrupole–octupole alignment with the dipole direction, coincident bulk-flow and quasar-polarization axes (Hutsemékers et al. 2005) — is motivation for H2 (Vortex)/H3 (Neighbour), not support.

Galaxy handedness. Claimed spin-handedness dipoles (Longo 2011; Shamir 2012, 2022) lie 70ř from each other and 50ř–74ř from the axes above; we treat them as systematics-limited (Iye et al. 2021). The JWST/JADES handedness claim 8.6ř from the Cold Spot is a coincidence awaiting a control-field test (Appendix A).

Falsifier. Absence of the E-mode ring at CMB-S4 sensitivity eliminates H2 (Vortex). A localized curl detection at the Cold Spot, P8 (Curl Test), would support the vortex reading over the neighbour reading of the same direction (Sec. 7), which is mutually exclusive with it; a calibrated curl null retires it (Table 2).

7. H3 (Neighbour) and H4 (Population): A Neighbouring Big Pool Bang in the Cold Spot Direction, and a Population of Neighbours

Because vacuum crystallization is stochastic across an unbounded background, our bang need not be unique. A neighbouring universe nucleating beyond our domain boundary imprints our last-scattering surface, at comoving distance $d_{\rm LS} \approx 46$ Gly (4.35×10^{26} m); a neighbour closer than that would appear in large-scale-structure tracers, which the Cold Spot sightline does not show beyond the too-shallow Eridanus supervoid.

Causal admissibility. A source at $D = (1+\epsilon)\, d_{\rm LS}$ with $\epsilon > 0$ lies beyond our particle horizon. Nothing emitted by it since the bounce — no black holes crossing the boundary, no gravitational-wave memory, no tidal torque acting over the past Gyr — can have reached our volume. The only causally admissible imprint is one already present in the initial conditions: a potential Φ of the Grishchuk–Zel'dovich type laid down at the bounce, which the last-scattering photons sample as a Sachs–Wolfe spot with a $1/r$ tail over the whole sky at low multipoles. The neighbour reading therefore makes two kinds of prediction: the spot itself (amplitude, profile, non-rotation, polarization edge) and its low-ℓ tail, P9 (Low-ℓ Tail). Its present-day effect on our volume is a static field — an induced (monopole) velocity of the local volume and a tidal shear — fixed once the mass and distance are (Sec. 8).

The Sachs–Wolfe model. For a potential Φ present at last scattering, $\Delta T/T = \alpha\, \Phi/c^2$ (Sachs & Wolfe 1967), where the coefficient α depends on how the perturbation was laid down: $\alpha = 1/3$ for adiabatic growing modes and $\alpha \to 2$ for isocurvature-like modes in which the potential is imposed without a compensating density perturbation. An externally imposed potential lies between these limits; we adopt $\alpha = 1$ and carry the resulting factor-of-six range — asymmetric about the adopted value, $M \times 3$ at $\alpha = 1/3$ and $M/2$ at $\alpha = 2$, i.e. $+0.5/-0.3$ dex — into the mass uncertainty of Sec. 8. At the spot centre $\Phi = GM/(\epsilon\, d_{\rm LS})$, and the angular profile is

$$\Delta T(\theta) \propto \Big[\epsilon^2 + 2(1+\epsilon)(1-\cos\theta)\Big]^{-1/2},$$

so the half-maximum angle fixes the distance, $\epsilon = \theta_{1/2}/\sqrt{3}$, and the amplitude then fixes the mass, $M = (\Delta T/T)\, \epsilon\, d_{\rm LS}\, c^2/(\alpha G)$. Because $\theta_{1/2}$ is beam- and smoothing-limited at 2ř smoothing, the inferred ϵ values are upper bounds, and since $M \propto \epsilon\, \Delta T$, the mass at fixed amplitude is an upper bound too. Amplitudes and widths must be paired at the same smoothing; we use the 2ř values (ΔT, $\theta_{1/2}$) throughout. The mass function of a population of such spots is the nucleation size spectrum of the vacuum transition — the distribution that drove the cascade of Sec. 1 — so the internal (ejecta) and external (spot) samples of one process must agree, a cross-consistency requirement linking H1 (Bounce) and H4 (Population).

Southern one-sidedness. All eight most-extreme large-scale spots on the SMICA map ($|b| > 25$ř, 2ř smoothing; Appendix A) lie in the southern galactic hemisphere; in 200 Gaussian realizations through the same pipeline none placed its full top-8 in one hemisphere (mock mean 5.2 of 8). The significance to be quoted is that of the published hemispherical power asymmetry ($p \sim$ 1–10%, estimator-dependent; Planck Collaboration 2020 VII), since our mocks are simplified and the frame post hoc. What this framework can say is limited: the Cold Spot and the one-sided extremes share a hemisphere, and a neighbour to the galactic south would add a few-μK potential tail on that side of the sky (Sec. 9d). Whether such an additive tail can produce the observed $\sim$7% hemispherical *power* modulation, or additional $\pm 150\,\mu$K extremes, has not been computed and is not claimed; a few-μK additive term does not obviously do either. The shared hemisphere is therefore a coincidence to be explained, not a prediction of H3 (Neighbour). An isotropic swarm of neighbours (H4 (Population)) predicts no such one-sidedness.

H4 (Population). Since nucleation is stochastic, the neighbour should not be unique: a population of Big Pool Bangs at varying distances, each imprinting a distinct anomaly, P6 (Spot Population). Planck 2018 isotropy analyses (Planck Collaboration 2020 VII) find the sky consistent with Gaussian ΛCDM apart from 2–3σ large-scale anomalies and the single Cold Spot, so additional neighbours must imprint below Planck's per-spot threshold. H4 (Population) stands or falls on population statistics (Sec. 9), not on any single spot.

Falsifiers. H3 (Neighbour) falls to a foreground structure accounting for the full decrement, a curl detection at the Cold Spot (which transfers the direction to H2 (Vortex)), or a P9 (Low-ℓ Tail) null at the pre-registered threshold; H4 (Population) to a null in sub-threshold spot statistics at LiteBIRD/CMB-S4 depth (Table 2).

8. H6 (Neighbour Location): Where Is the Second Big Pool Bang?

H6 (Neighbour Location) states the neighbour's location once, summarized in Table 3; other sections refer here.

Direction. The Cold Spot, $(l, b) = (203ř, -56ř)$, uncertain by $\sim$2–3ř (the half-max radius), in the hemisphere of the eight-spot one-sidedness.

Distance. With $\theta_{1/2} = 2.5ř$ the profile of Sec. 7 gives $\epsilon = 0.025$; the eight extremes span $\epsilon = 0.025$–0.055 (Sec. 9). The source lies $\epsilon\, d_{\rm LS} \approx 1.1$ Gly comoving beyond the last-scattering surface (up to 2.5 Gly for the widest spots), at $D \approx 47$ Gly $\approx 4.5 \times 10^{26}$ m; since $\theta_{1/2}$ is smoothing-limited, ϵ is an upper bound.

Mass. With $\Delta T = -168\,\mu$K and $\theta_{1/2} = 2.5ř$ (both at 2ř smoothing), $T = 2.725$ K, $\epsilon = 0.025$, Sachs–Wolfe coefficient $\alpha = 1$ and $d_{\rm LS} = 4.35 \times 10^{26}$ m,

$$M_2 = \frac{\Delta T}{T}\,\frac{\epsilon\, d_{\rm LS}\, c^2}{\alpha G} = (6.2 \times 10^{-5})\,(1.09 \times 10^{25}\ \mathrm{m})\,(1.35 \times 10^{27}\ \mathrm{kg\ m^{-1}}) \approx 9 \times 10^{47}\ \mathrm{kg} \approx 10^{48}\ \mathrm{kg},$$

uncertain by $+0.6/-0.4$ dex: $+0.5/-0.3$ dex from the Sachs–Wolfe coefficient ($M \times 3$ at $\alpha = 1/3$, $M/2$ at $\alpha = 2$; Sec. 7) and ± 0.3 dex from the range of ϵ across the population (0.025–0.055), added in quadrature. Because ϵ is an upper bound, M_2 at fixed ΔT is an upper bound. The result is $\sim 4 \times 10^{-5}$ of our own per-Hubble-volume $M_1 = 2.2 \times 10^{52}$ kg, a dwarf neighbour. We use $M_2 \approx 10^{48}$ kg ($+0.6/-0.4$ dex) throughout.

Induced velocity of the local volume. The neighbour's present-day acceleration of our volume is $GM_2/D^2 = (6.67 \times 10^{-11})(9 \times 10^{47})/(4.5 \times 10^{26})^2 \approx 3 \times 10^{-16}$ m s^{-2}; integrated over a Hubble time (4.4×10^{17} s) this is ≈ 130 m s^{-1}, the induced (monopole) velocity of the local volume — ~ 0.1 km s^{-1}, and $\lesssim 1$ km s^{-1} after the mass uncertainty. This is negligible against the 428 ± 36 km s^{-1} Cosmicflows-4 bulk flow at $200\,h^{-1}$ Mpc (Watkins et al. 2023) toward (298ř, −8ř), 86ř from the Cold Spot. The neighbour reading therefore predicts *no* bulk-flow contribution above ~ 1 km s^{-1}, stated as P10 (Induced-Velocity Bound); the observed flow must be explained by local attractors. The genuinely tidal part is the shear: a tidal tensor $GM_2/D^3 \approx 7 \times 10^{-43}$ s$^{-2} \approx 1.5 \times 10^{-7}\,H_0^2$ (with $H_0^2 \approx 4.8 \times 10^{-36}$ s^{-2}), which integrated over a Hubble time gives a dimensionless expansion-rate anisotropy $\delta H/H_0 \sim 10^{-7}$, undetectable in supernova anisotropy searches, far below the limits on anisotropic expansion (Saadeh et al. 2016) and below the low-ℓ Grishchuk–Zel'dovich (1978) limit — a consistency check, not a discriminant. Hot spots require a boundary imprint of the opposite sign (an underdense or approaching wall); the model does not specify which neighbours imprint hot.

Our position relative to it. ≈ 47 Gly comoving along the axis toward (203ř, −56ř); the internal offset of H5 (Our Position), $\lesssim$ few Mpc (Sec. 10), is negligible by comparison.

Table 3. H6 (Neighbour Location): the neighbour's direction, distance, mass and local effects.

Quantity	Estimate	Method	Status
Direction	(203ř, −56ř) ± 3ř	Cold Spot centroid; same hemisphere as the eight extremes	Fixed by data; attribution hypothetical
Distance beyond last scattering	$\epsilon = 0.025$, ≈ 1.1 Gly comoving (upper bound)	$\epsilon = \theta_{1/2}/\sqrt{3}$ at 2ř smoothing	Smoothing-limited
Mass M_2	$\approx 10^{48}$ kg ($+0.6/-0.4$ dex; upper bound at fixed ΔT)	Sachs–Wolfe, coefficient $\alpha = 1$ (1/3–2), $\Delta T = -168\,\mu$K at 2ř	Single adopted value
Induced local velocity (P10 (Induced-Velocity Bound))	$\lesssim 1$ km s^{-1}	$GM_2/D^2 \times t_H$ (monopole)	Bound; flow is local
Tidal tensor	$\sim 1.5 \times 10^{-7}\,H_0^2$ (expansion-rate anisotropy $\delta H/H_0 \sim 10^{-7}$)	GM_2/D^3	Consistent, undetectable
Curl at Spot (P8 (Curl Test))	Zero (neighbour) vs excess (vortex)	Lensing curl / B-mode stack	Uncalibrated null; calibrated test pending
Low-ℓ tail (P9 (Low-ℓ Tail))	Fixed $\ell = 2, 3$ template	Correlation vs Gaussian null (Sec. 9)	Uninformative at present

What H6 (Neighbour Location) predicts. A *non-rotating* imprint (zero curl at calibrated sensitivity), a disc-shaped edge with the Aguirre & Johnson (2011) bubble-collision polarization profile, a low-ℓ tail of fixed shape (P9 (Low-ℓ Tail)), no bulk-flow contribution (P10 (Induced-Velocity Bound)), and no proper motion. It does not predict the hemispherical power asymmetry (Sec. 7).

Falsifiers are in Table 2 (Sec. 11); P8 (Curl Test) is the decisive near-term test, separating the two readings of the same direction.

9. H7 (Pool Model): The Population

(a) Universe 1 is one of several. Vacuum crystallization nucleated more than once; our bang is one of several separated by domain boundaries, each with its own furnace and ejected halo.

(b) Each neighbour imprints one spot. Under the Sachs–Wolfe model of Sec. 7, the eight most extreme large-scale spots on the SMICA map are the candidate neighbour list (Table 4), each with $\epsilon = \theta_{1/2}/\sqrt{3}$ (an upper bound, since $\theta_{1/2}$ is smoothing-limited at 2ř; amplitudes and widths are both the 2ř-smoothing values, and each implied mass carries the $+0.6/-0.4$ dex of Sec. 8):

Table 4. The eight most extreme large-scale spots of the Planck SMICA map ($|b| > 25$ř, 2ř smoothing).

Rank	(l, b)	Sign	ΔT (μK)	$\theta_{1/2}$	ϵ (upper bound)
1	(157ř, -71ř)	hot	+170	2.5ř	0.025
2	(80ř, -33ř)	cold	-170	3.5ř	0.035
3	(203ř, -56ř) (Cold Spot)	cold	-168	2.5ř	0.025
4	(155ř, -29ř)	cold	-163	5.5ř	0.055
5	(305ř, -29ř)	hot	+153	2.5ř	0.025
6	(211ř, -35ř)	cold	-153	2.5ř	0.025
7	(170ř, -47ř)	hot	+162	4.5ř	0.045
8	(184ř, -54ř)	hot	+155	4.5ř	0.045

The eight extremes require $\epsilon = 0.025$–0.055, i.e. sources 1–2.5 Gly (comoving) beyond last scattering. Under Sachs–Wolfe an overdense wall gives a *cold* spot; a hot spot requires an underdensity or an approaching (Doppler) wall, a sign mechanism the model leaves unspecified (Sec. 8). Spot 8 at (184.2ř, -54.3ř) lies 11ř from the Cold Spot centroid (203.2ř, -56.3ř) and may be part of one system; spot 7 at (170.3ř, -46.6ř) lies 22ř away and is separate. The candidate count is ≤ 8, or 7 if spot 8 is merged.

(c) Population predictions. Neighbours cluster on the sky (currently 8/8 south). Any genuine neighbour is non-rotating and should show zero curl; any relic vortex of our own bang, H2 (Vortex), should show a curl excess under P8 (Curl Test) — the population divides on this test. The inferred neighbour mass function must match the nucleation spectrum inferred from our own ejecta. The exploratory locus, curl, lensing-convergence and tracer statistics on the eight spots are reported once, in Appendix A; none is significant.

(d) The low-multipole tail, P9 (Low-ℓ Tail). A neighbour that imprints a compact spot cannot imprint *only* a spot: the $1/r$ potential of each source extends over the whole sky at $\sim$2% of its peak (3–6 μK at 90ř, 2–4 μK at the antipode). Summing the eight tails gives a fixed $\ell = 1$–3 template with rms 4.1 μK at $\ell = 2$ and 4.5 μK at $\ell = 3$, i.e. $\sim$20% of the observed quadrupole power ($C_2 \approx 210\,\mu\mathrm{K}^2$) and $\sim$8% of the octopole — allowed, and a prediction: the template must correlate positively with the observed $\ell = 2, 3$ maps. *Pre-registered threshold.* With only 5 ($\ell = 2$) and 7 ($\ell = 3$) real modes per multipole, the correlation coefficient r between independent Gaussian patterns has $\sigma \approx 0.4$–0.5 under the null, so a single value of r is uninformative. The test is: compute r between the template and the masked Planck Commander $\ell \leq 3$ maps (Planck Collaboration 2020 IV), and the same r

for 1000 Gaussian ΛCDM realizations (HEALPix synfast, same mask and pipeline); a detection requires r above the 95th percentile of the null at *both* multipoles, and a value at or below the null median at both, with the template amplitude fixed by the spots, falsifies the population reading. An unmasked SMICA comparison ($r = +0.15$ at $\ell = 2$, -0.27 at $\ell = 3$; template axes 78ř and 52ř from the observed) is uninformative at this power and foreground-contaminated.

(e) One-sided or surrounding. A *surrounding* population would populate both hemispheres; the 8/8 southern count indicates a one-sided, clustered population, and H7 (Pool Model) takes that as its picture. The symmetric lattice of Appendix B is the limiting symmetric case, whose cubic-harmonic ($\ell = 4$) expansion-rate signature is a further falsifier. The pre-registered extension is the hemispheric balance of the extremes ranked 9–28: a one-sided population predicts a persistent southern excess; a surrounding population, or a Gaussian sky with the top-8 count a fluctuation, predicts isotropy. What falsifies H7 (Pool Model) as a whole is given in Table 2 (Sec. 11).

10. H5 (Our Position): Where Are We Relative to Our Own Big Pool Bang?

H5 (Our Position) is the discriminator between the two geometries of Sec. 3. In Geometry A ejection happens everywhere and there is no geometric centre, so H5 (Our Position) predicts a null. In Geometry B the rebound was a centred event and radial ordering survives, so H5 (Our Position) asks: **does any residual large-scale radial ordering survive in our Hubble volume?** Two signatures would answer yes: a displaced-centre dipole $\boldsymbol{\xi}$ in the structure-subtracted peculiar-velocity field above the mock noise floor defined below, or a radial-ordering signature in the bulk-flow profile (an amplitude that grows, rather than decays, with survey depth). Either supports B; an offset consistent with the mock distribution together with a decaying flow favours A. Results are summarized in Table 5.

Displacement from the centre. The machinery is standard (King & Ellis 1973; Alnes & Amarzguioui 2006; Asvesta et al. 2022). Attributing the full CMB dipole (370 km s^{-1}) to off-centring gives $\Delta r \approx 5$ Mpc; the residual unexplained by local gravity gives $\Delta r \lesssim 1$–2 Mpc. A displaced centre is invisible in a purely linear flow and enters only through the curvature of the velocity law, at second order, $\sim H|\boldsymbol{\xi}|^2/2r$: ≈ 90 km s^{-1} at $r = 10$ Mpc for a 5 Mpc offset. Fitting the 2{,}112 Cosmicflows-4 groups (Tully et al. 2023) at 1.5–40 Mpc with $v(R) = hR + kR^2$ diverging from a free centre gives an offset $|\boldsymbol{\xi}| = 6.1$ Mpc toward (118ř, −24ř) at $\Delta\chi^2 = 6595$, with shells at 40–80 and 80–150 Mpc returning centres within 10ř–15ř of it. Two controls change the picture. *Mock mask:* ten isotropic mocks on CF4's exact sky positions and distance errors manufacture false centres of $|\boldsymbol{\xi}| = 0.4$–2.5 Mpc ($\Delta\chi^2 \approx 9$–61 per shell) that in the outer shells point into the *same longitude band* ($l \approx$ 93ř–152ř) as the data. We define the **mock floor** as the 95th percentile of the structure-subtracted mock $|\boldsymbol{\xi}|$ distribution; with the present ten mocks that percentile is set by the largest false centres, so $|\boldsymbol{\xi}|_{\text{floor}} \approx 2.5$ Mpc, and a tighter value awaits a larger mock ensemble. *Structure subtraction:* removing the 2M++ reconstruction-predicted peculiar velocities (Carrick et al. 2015) drops the inner shell to $\Delta\chi^2 = 182$, $|\boldsymbol{\xi}| = 1.7$ Mpc, within 8ř of the CMB dipole, and the outer shells into the mock range. The 1.7 Mpc residual lies *inside* the mock distribution (0.4–2.5 Mpc) and is therefore not a detection by the criterion just stated. What makes it worth one more test rather than immediate dismissal is its direction, which is that of the external dipole the 2M++ reconstruction itself calibrates against the CMB frame; whether that direction is signal or calibration is decided by one re-run: repeating the structure-subtracted fit with the 2M++ external dipole left free as a nuisance vector. If $|\boldsymbol{\xi}|$ stays at $\sim$1.7 Mpc with the same direction, it is a calibration-independent residual that still sits inside the mock distribution and must wait for the deeper catalogue; if it collapses, it was the calibration. Pending that re-run, **no displaced**

expansion centre is claimed once mapped structure is subtracted; the current bound is $|\boldsymbol{\xi}| \lesssim 2$ Mpc, $\sim 10^{-4}$ of the distance to last scattering.

The bulk flow. The bias-corrected Cosmicflows-4 bulk flow (Watkins et al. 2023) is 428 ± 36 km s^{-1} at $200\,h^{-1}$ Mpc toward (298ř $\pm$ 5ř, $-$8ř $\pm$ 4ř), rising with depth, in nominal $\sim 4.8\sigma$ tension with ΛCDM (Whitford et al. 2023 argue the errors are understated). Planck's kSZ analysis (Planck Collaboration 2014 XIII) limits coherent flows to $\lesssim 254$ km s^{-1} at 95%, but the two numbers refer to different scales: the kSZ limit applies to the coherent motion of clusters at Gpc depth, whereas the CF4 flow is a shallower measurement within $200\,h^{-1}$ Mpc, inside which a flow of several hundred km s^{-1} sourced by mapped attractors is not in conflict with it. What the kSZ limit disfavors is a flow of $\gtrsim 250$ km s^{-1} *persisting to Gpc depth*, the second half of P4 (Off-Centre Flow). The flow axis and the CMB dipole (264ř, +48ř) lie 63ř apart, which a single offset vector does not predict; local-attractor infall explains the flow without any offset, and the neighbour of Sec. 8 contributes $\lesssim 1$ km s^{-1}. The Cold Spot direction is 86ř from the flow axis and 116ř from the CMB dipole, so the southern one-sidedness (Sec. 7) is not an offset signature either.

Table 5. H5 (Our Position): offset and bulk-flow status.

Quantity	Estimate	Method	Status
Offset from centre $\lvert\boldsymbol{\xi}\rvert$	$\lesssim 2$ Mpc; 1.7 Mpc structure-subtracted residual, inside the mock distribution	CMB dipole cap; CF4 free-centre fit with mock-mask and 2M++ controls	No detection claimed; free-dipole re-run pending
Mock floor	≈ 2.5 Mpc: 95th percentile of the structure-subtracted mock $\lvert\boldsymbol{\xi}\rvert$ distribution (mocks give 0.4–2.5 Mpc)	Ten isotropic mocks on the CF4 mask, $\Delta\chi^2 \approx 9$–61	Defines the support threshold
Offset direction	Undetermined (CMB dipole (264ř, +48ř) vs flow (298ř, $-$8ř))	Dipole/flow attribution	Misaligned by 63ř
Bulk flow (P4 (Off-Centre Flow))	428 ± 36 km s^{-1} at $200\,h^{-1}$ Mpc; $\gtrsim 250$ km s^{-1} persisting to Gpc depth disfavored	Cosmicflows-4, Planck kSZ	Growth with depth untested beyond $200\,h^{-1}$ Mpc

Support and falsification, in one unit. Both criteria are stated in $|\boldsymbol{\xi}|$ against the mock distribution. *Support for H5 (Geometry B):* a structure-subtracted $|\boldsymbol{\xi}|$ above the 95th-percentile mock floor (≈ 2.5 Mpc for the present catalogue), obtained with the 2M++ external dipole left free, with a direction stable across shells; and/or a bulk-flow amplitude that grows with depth beyond $200\,h^{-1}$ Mpc (P4 (Off-Centre Flow)). *Falsification of Geometry B:* a structure-subtracted, free-dipole $|\boldsymbol{\xi}|$ consistent with the mock distribution in the next Cosmicflows release, together with a bulk flow that decays with depth. The current status is neither: $|\boldsymbol{\xi}| \lesssim 2$ Mpc, with the 1.7 Mpc residual inside the mock distribution and the free-dipole re-run pending.[3]

[3] *Status and retirement note (distinct from falsification).* If the next Cosmicflows release, with roughly twice the group count and a $\sim\sqrt{2}$ smaller offset error, returns a structure-subtracted, free-dipole $|\boldsymbol{\xi}|$ consistent with its own

11. Falsifiable Predictions, Instrumentation, and Falsification Criteria

Hypothesis hierarchy. H1 (Bounce; P3 (Transition GW Background), P5 (No Particle DM)) is the core and must hold; H2 (Vortex; P1 (Polarization Ring), P2 (Cyclonic B-modes), P7 (Spin Surcharge), P8 (Curl Test)), H3 (Neighbour; P8 (Curl Test), P9 (Low-ℓ Tail)), H4 (Population; P6 (Spot Population)), H5 (Our Position; P4 (Off-Centre Flow)), H6 (Neighbour Location; P8 (Curl Test), P9 (Low-ℓ Tail), P10 (Induced-Velocity Bound)) and H7 (Pool Model; P6 (Spot Population), P9 (Low-ℓ Tail)) are separable from the core: any can fail without falsifying the core mechanism. They are not all mutually independent, however. H2 (Vortex) and H3 (Neighbour) are *mutually exclusive* readings of the Cold Spot direction — the Spot is a relic vortex of our own bang or a neighbour's non-rotating imprint, not both — and the curl test P8 (Curl Test) decides between them: a localized curl excess transfers the direction to H2 (Vortex), a calibrated null to H3 (Neighbour). H6 (Neighbour Location) stands or falls with H3 (Neighbour), and H7 (Pool Model) presupposes H4 (Population). The remaining pairs can hold or fail independently.

Table 1. Predictions.

#	Prediction (hypothesis)	Observable and required threshold	Instrument	Status
P1 (Polarization Ring)	Cold Spot ring (H2 (Vortex)). Thermal ring of $+20$–$50\,\mu$K at 5ř–7ř: **failed** in Planck 2018 (Sec. 6). Surviving prediction: radial E-mode ring of 1–3 μK at 5ř–15ř	Polarization sensitivity $\lesssim 1\,\mu\text{K}\cdot\text{arcmin}$ at $\ell \sim 20$–100	CMB-S4	Thermal: failed. Polarization: open
P2 (Cyclonic B-modes)	Cyclonic B-mode pattern aligned with the Axis of Evil (H2 (Vortex))	Degree-scale B-modes, r-equivalent $\sim 10^{-3}$	CMB-S4, LiteBIRD	Open
P3 (Transition GW Background)	First-order-transition stochastic gravitational-wave background from bubble collisions in the crystallization (H1 (Bounce)): broken power law peaked at the transition scale	Stochastic-background spectral shape at nHz–mHz	Pulsar timing arrays, LISA	Open

(correspondingly tighter) mock distribution, but the bulk flow neither grows nor clearly decays, then H5 (Our Position) has no observable signature at distance-catalogue depth and is *retired as untestable at present*. Retirement is a statement about instrument reach, not a falsification of Geometry B, and is kept out of Table 2's falsifier column for that reason.

#	Prediction (hypothesis)	Observable and required threshold	Instrument	Status
P4 (Off-Centre Flow)	Radial velocity ordering in our Hubble volume — Geometry B's own prediction (H5 (Our Position)): a structure-subtracted off-centre dipole $\lvert\boldsymbol{\xi}\rvert$ above the 95th-percentile mock floor (≈ 2.5 Mpc at present, Sec. 10) with the 2M++ external dipole free and a direction stable across shells, and/or a bulk-flow amplitude that *grows* with survey depth beyond $200\,h^{-1}$ Mpc (currently 428 ± 36 km s^{-1} toward $l \approx 298$ř). Geometry A predicts a null: $\lvert\boldsymbol{\xi}\rvert$ consistent with the mock distribution and a decaying flow. A flow $\gtrsim 250$ km s^{-1} persisting to Gpc depth is disfavored by the Planck kSZ limit	Next Cosmicflows release (free-dipole re-run); peculiar-velocity survey to $z \sim 1$ with $\lesssim 300$ km s^{-1} per cluster	CF4 (archival re-run); Rubin/LSST + Euclid + kSZ	$\lvert\boldsymbol{\xi}\rvert \lesssim 2$ Mpc, 1.7 Mpc residual inside the mock distribution; flow rising to $200\,h^{-1}$ Mpc, errors disputed
P5 (No Particle DM)	No particle dark-matter candidate (DM = Planck relics) (H1 (Bounce))	Continued nulls approaching the neutrino floor	LZ, XENONnT	Ongoing

#	Prediction (hypothesis)	Observable and required threshold	Instrument	Status
P6 (Spot Population)	A sub-threshold population of Cold-Spot-class spots (H4 (Population); H7 (Pool Model)); Planck already limits any above-threshold population	Sub-threshold spot statistics; per-candidate polarization at $\lesssim 1\,\mu\mathrm{K}\cdot\mathrm{arcmin}$; hemispheric balance of extremes 9–28	LiteBIRD, CMB-S4; Planck (archival)	Constrained; open below threshold
P7 (Spin Surcharge)	Spin surcharge (H2 (Vortex)): filaments and clusters carry primordial angular momentum atop tidal spin, excess largest at high z, randomly oriented (no net spin above shot noise; $\omega/H_0 \lesssim 10^{-9}$; cf. the lattice-spin cap of Sec. 12). The Wang et al. (2021) filament spin is the $z \approx 0$ anchor	Filament-spin stacking at $z \sim 1$ vs $z \sim 0$ against ΛCDM torque mocks	DESI, Euclid, 4MOST	Open

#	Prediction (hypothesis)	Observable and required threshold	Instrument	Status
P8 (Curl Test)	Localized curl excess (lensing curl and B-mode variance) at the extreme-spot positions if they are relic vortices (H2 (Vortex)); zero at the Cold Spot if it is a neighbour imprint (H3 (Neighbour); H6 (Neighbour Location)). A calibrated stacked null at the Cold Spot retires the vortex reading (Appendix A)	Planck PR4 curl reconstruction at full depth; degree-scale B-modes at $\lesssim 1\,\mu\mathrm{K}\cdot\mathrm{arcmin}$	Planck PR4 (archival), CMB-S4, LiteBIRD	Uncalibrated null; calibrated test pending
P9 (Low-ℓ Tail)	Low-multipole tail (H3 (Neighbour); H6 (Neighbour Location); H7 (Pool Model)): the fixed $\ell = 1$–3 template summed from the eight point-mass profiles (rms $4\,\mu$K at $\ell = 2$, $4.5\,\mu$K at $\ell = 3$) must correlate positively with the observed $\ell = 2, 3$ maps; detection requires r above the 95th percentile of 1000 Gaussian realizations at both multipoles (Sec. 9d)	Masked Commander low-ℓ maps (Planck Collaboration 2020 IV); r against the simulated null	Planck 2018/PR4 (archival)	Unmasked $r = +0.15$ ($\ell = 2$), -0.27 ($\ell = 3$): uninformative

#	Prediction (hypothesis)	Observable and required threshold	Instrument	Status
P10 (Induced-Velocity Bound)	Induced-velocity bound (H6 (Neighbour Location)): the neighbour's induced (monopole) velocity of the local volume is $\lesssim 1$ km s^{-1} (Sec. 8), so it contributes nothing measurable to the bulk flow; the observed 428 km s^{-1} must be accounted for by local attractors	Bulk-flow residual after 2M++/CF4 reconstruction $\ll 400$ km s^{-1} unexplained	CF4 (archival); DESI peculiar velocities	Bound; consistent

Table 2. Falsification criteria. This is the single reference for what each hypothesis needs and what removes it; Secs. 9, 12 and the Epilogue refer here.

Hypothesis	What supports it	What falsifies it	Current status
H1 (Bounce) bounce and ejection	$88\% \pm 4\%$ ejected fraction at $\sigma_{\rm rel} \approx 1$; particle-DM nulls (P5 (No Particle DM)); a transition-shaped GW background (P3 (Transition GW Background))	A particle dark-matter detection; an ejecta velocity spectrum that cannot yield a cold matter component by matter–radiation equality; once a derivation of ρ_c from the jamming limit exists, a derived value differing from the LQC $0.41\,\rho_P$ by more than a factor of 3	Open; relativistic formulation and ejecta temperature unresolved

Hypothesis	What supports it	What falsifies it	Current status
H2 (Vortex) Cold Spot vortex	E-mode ring (P1 (Polarization Ring)), cyclonic B-modes (P2 (Cyclonic B-modes)), curl excess at the Spot (P8 (Curl Test)), spin surcharge (P7 (Spin Surcharge))	Absence of the E-mode ring at CMB-S4 sensitivity; a calibrated stacked curl null at the Cold Spot	Thermal ring failed; polarization open; curl uncalibrated
H3 (Neighbour) neighbour in the Cold Spot direction	Zero curl at the Spot (P8 (Curl Test)); positive low-ℓ tail correlation (P9 (Low-ℓ Tail)); bubble-collision polarization edge	A foreground structure accounting for the full decrement; a curl detection at the Spot; P9 (Low-ℓ Tail) at or below the null median at both multipoles	Open; P9 (Low-ℓ Tail) uninformative
H4 (Population) population of neighbours	Sub-threshold spot population (P6 (Spot Population)); mass function matching the ejecta nucleation spectrum	Null sub-threshold spot statistics at LiteBIRD/CMB-S4 depth	Constrained above threshold; open below
H5 (Our Position) residual radial ordering (Geometry B vs A)	Structure-subtracted, free-dipole $\lvert\boldsymbol{\xi}\rvert$ above the 95th-percentile mock floor (≈ 2.5 Mpc at present) with a direction stable across shells; bulk flow growing with depth (P4 (Off-Centre Flow)). Either supports Geometry B	Structure-subtracted, free-dipole $\lvert\boldsymbol{\xi}\rvert$ consistent with the mock distribution in the next Cosmicflows release, together with a bulk flow decaying with depth: falsifies Geometry B (retirement as untestable is a separate status, Sec. 10 footnote)	Unresolved; $\lvert\boldsymbol{\xi}\rvert \lesssim 2$ Mpc, 1.7 Mpc residual inside the mock distribution, awaiting the free-dipole re-run
H6 (Neighbour Location) location of the second bang	Non-rotating imprint (P8 (Curl Test)); low-ℓ tail (P9 (Low-ℓ Tail)); no induced-velocity contribution to the bulk flow (P10 (Induced-Velocity Bound)); Aguirre–Johnson polarization edge	Curl detection at the Spot; absence of the bubble-collision polarization profile at CMB-S4 sensitivity	Open

Hypothesis	What supports it	What falsifies it	Current status
H7 (Pool Model) one-sided population	Southern excess persisting in extremes 9–28 (P6 (Spot Population)); positive P9 (Low-ℓ Tail); zero curl at all eight spots	All of: P9 (Low-ℓ Tail) failing at the pre-registered threshold; zero curl at calibrated sensitivity at all eight spots *and* no bubble-collision polarization profile at any at CMB-S4 depth; no CMB-independent mass counterpart in a pre-registered galaxy-density re-run. For the lattice limit: a detected $\ell = 4$ cubic expansion-rate pattern (Appendix B)	Open; 8/8 south at $p \sim 1$–10%

Under the joint nulls of H3 (Neighbour), H4 (Population), H6 (Neighbour Location) and H7 (Pool Model) the spot population is Gaussian and the Cold Spot a single outlier of unknown origin, while H1 (Bounce) and H2 (Vortex) stand and H5 (Our Position) continues to arbitrate between Geometries A and B. The polarization ring and calibrated curl test await Simons Observatory / CMB-S4 (a 2030s timeline); P9 (Low-ℓ Tail) and P10 (Induced-Velocity Bound) are archival.

12. Limitations and Open Problems

1. **Hard-sphere ansatz.** Treating Planck relics as incompressible spheres is a modeling assumption motivated by the Compton–Schwarzschild coincidence, not a quantum-gravity result.
2. **Bounce mechanism and horizon structure.** The rebound requires the LQC-form modified Friedmann equation (Bojowald 2001; Ashtekar & Singh 2011); $\rho_c = \eta_{\rm rcp}\rho_P$ is a conjecture. The bounce is meaningful without qualification only for a whole or super-horizon patch (Geometry A). The finite-cloud Geometry B *currently has no bounce mechanism*: a finite cloud of the relevant mass is $\sim 10^{40}$ inside its Schwarzschild radius, the homogeneous LQC equation offers it no exit, and no trapped-surface timing analysis — which would also have to accommodate the multi-bubble assembly timescale ($\gtrsim 10^{20}\, t_P$ per Hubble volume) — has been done. Geometry B is retained only as the geometry H5 (Our Position) can test kinematically.
3. **Spectral index.** ν_t and the relation $n_s - 1 = -\nu_t/3$ are fitted inputs; the compatibility of a Kolmogorov cascade with near scale invariance is unproven.
4. **Dark-matter fraction and temperature.** The $88\% \pm 4\%$ ejection requires $\sigma_{\rm rel} \approx 1$, which the cascade model reaches only marginally, and the end-to-end chain is untested. The ejecta are relativistic, hence hot rather than cold dark matter unless redshifting or cooling reduces their free-streaming length by matter–radiation

equality. In Geometry A this is also the condition for the ejection-set peculiar velocities to redshift into the Hubble flow at all (Secs. 3, 5.2). Planck-relic dark matter must also face femtolensing, capture, and abundance constraints.

5. **Expansion history.** Neither the Newtonian toy nor the LTB lightcone produces apparent acceleration, the toy's $w(z)$ drift runs opposite to the DES+DESI direction, and the kSZ gate caps any ejecta contribution at $\sim 1\%$ of dark energy. The adopted history (Sec. 5.2) is the Big Pool Bang framework plus a constant tension on an assumed flat ΛCDM background (Geometry A); the tension's value is not predicted, and Geometry B's own monopole distance–redshift relation (the LTB-plus-tension lightcone) has not been computed. The model is also not yet reconciled with BBN abundances or the CMB acoustic peaks.
6. **Causal and relativistic structure.** At the bounce the causal patch is of Planck size and one present-day Hubble volume comprises $\sim 10^{60}$ of them; homogeneity across causal patches must be assumed (the horizon problem stands), and in Geometry A, where the jammed patch is super-horizon, its total extent is unconstrained. Ejection proceeds where Newtonian escape velocity vastly exceeds c; the N-body mechanism lacks a relativistic formulation, prerequisite for H1 (Bounce) to be a physical theory rather than an analogy. Which of the two geometries of Sec. 3 applies is itself undetermined by the theory and is left to H5 (Our Position). The neighbour of H3 (Neighbour)/H6 (Neighbour Location) lies beyond the particle horizon, so its imprint must be an initial-condition potential (Sec. 7); how such a potential is laid down across a domain boundary is not modelled.
7. **Relic-to-plasma conversion.** The furnace must convert relics to Standard-Model plasma with the observed baryon asymmetry while ejected relics stay stable; no channel is specified.
8. **Sign of hot spots and the Sachs–Wolfe coefficient.** Sachs–Wolfe supplies cold spots from overdense walls; the hot extremes require an unspecified underdensity or Doppler mechanism (Sec. 8). The coefficient relating Φ to $\Delta T/T$ for an externally imposed potential is bracketed (1/3–2, a factor of six) rather than derived, which is the dominant term in the $+0.6/-0.4$ dex on M_2.
9. **Flatness and one-sidedness.** Spatial flatness is not addressed by the energy budget of the rebound (Sec. 4). Whether a neighbour's few-μK potential tail can generate the observed hemispherical power modulation has not been computed (Sec. 7).

Each item maps onto the observational program of Sec. 11 (Tables 1 and 2); the theory is offered as a falsifiable mechanical alternative, not a completed replacement for ΛCDM.

Further research: spin and an equal-mass lattice. A bang born from an isotropic bounce carries no net spin above shot noise (the per-sky ejection dipole of Sec. 3 matches the shot-noise expectation); like a planet or a galaxy it acquires spin only through tidal torque from neighbours, at second order (rate $\sim q\,GM/D^3$ for a cloud of asymmetry q). If every lattice bang is fixed at our own mass per Hubble volume, $M_1 = 2.2 \times 10^{52}$ kg, two order-of-magnitude constraints follow that we have not developed here. First, the $1/r$ potential tail of an equal-mass neighbour keeps the CMB quadrupole below $\sim 4\,\mu$K only for spacings $D \gtrsim 10^3$ Gly ($\sim 20\,d_{\rm LS}$); at the Cold Spot distance an equal-mass neighbour would imprint a ~ 4 K spot, so under Geometry B the Cold Spot source is either a bang of order 10^{-5}–10^{-4} of our mass or not a bang at all. In Geometry B the cloud mass is at least the matter mass of the observable universe, $\gtrsim 1.5 \times 10^{53}$ kg, which raises both numbers: the equal-mass spot becomes $\gtrsim 28$ K and the quadrupole-safe spacing $D \gtrsim 2 \times 10^3$ Gly. Second, because $\omega \propto q\,GM/D^3$

and the quadrupole bound already fixes the minimum spacing, the tidally induced global spin of an equal-mass lattice is *capped*, at $\omega/H_0 \lesssim 10^{-8}\,(q/0.1)$ with shear $\sim 10^{-7}$. A spin detection therefore cannot calibrate the lattice from above: any detected global spin above that cap — at $10^{-6}\,H_0$, say — would *contradict* the equal-mass lattice rather than calibrate it, whereas a detection at or below the cap would fix the spacing. The relevant existing bound is that of Saadeh et al. (2016), who limit the vorticity of homogeneous Bianchi VII$_h$ modes to $(\omega/H_0)_0 \lesssim 10^{-10}$ from the Planck temperature and polarization quadrupole and spiral patterns. Whether that bound applies here is an open question that we state honestly rather than settle: it is derived for homogeneous *decaying* vorticity modes, whereas a tidally induced spin is a growing second-order mode; but its CMB signature is the same quadrupole/spiral pattern already bounded, and if the Saadeh bound transfers directly the equal-mass spacing rises to $D \gtrsim 5 \times 10^3$ Gly. Late-time velocity-field curl statistics reach, as an order-of-magnitude estimate, only $\omega/H_0 \sim 10^{-2}$ and add nothing. The connection to P7 (Spin Surcharge) is the $\omega/H_0 \lesssim 10^{-9}$ quoted there: the primordial spin surcharge of P7 (Spin Surcharge) is randomly oriented and averages to zero on large scales, whereas the lattice spin is a single global axis, and the two are separately bounded. A treatment of tidal-torque spin-up of a finite cloud in Geometry B, with the shear and spin axes and their CMB signatures, is left to a separate paper.

Data and code availability. Code, figures, build instructions and download links for all datasets used are at https://github.com/AIMFIRST-VN/big001.

Epilogue: Three Questions, Three Answers

We set out with three questions and a pool table. Here is where the balls came to rest.

How does it work? Picture the moment before the first moment: a false vacuum, tense as a drawn bow. Somewhere a bubble of true vacuum nucleates, its wall sweeping outward, catalysing more bubbles, until the whole patch is a jammed crush of Planck-mass spheres pressed to the random-close-packing limit — some 10^{60} of them in what will become each Hubble volume, and no end to the patch that we can see. Nothing can fall further. A loop-quantum bounce fires, the pile rebounds, and the rebound is chaotic: $88\% \pm 4\%$ of the spheres are hurled outward, sorted by speed like shot from a gun. That flying debris is our dark matter. What its speed-sorting means depends on which of two geometries the universe chose. In Geometry A the crush was everywhere and had no centre: the ejection set the matter's initial peculiar velocities on a homogeneous background, and those velocities redshift into the Hubble flow only if the cooling problem of Sec. 12 is solved. In Geometry B the rebound was one centred event: the fastest went furthest, and that ordering *is* a ballistic Hubble-like law, $v = r/t$, to which the tension below is added — at the price of having, as yet, no bounce mechanism at all (a finite cloud of that mass sits 10^{40} inside its own horizon, and nothing in the equations brings it out) and of a monopole distance–redshift relation that has not yet been computed. Either way the debris does not push the universe faster; nothing outside a volume ever can, and nothing inside it pushes unless it has negative pressure. The acceleration we measure comes from a constant tension of space, $w = -1$, which we adopt on an assumed ΛCDM background and cannot yet derive. So the model works as far as the matter goes, hands the expansion to a tension it does not explain, and carries three open problems: the ejecta are born relativistic and must somehow cool, the thermal ring around the Cold Spot is not there, and the bounce is only meaningful without qualification if the jammed patch was the whole universe rather than a ball inside its own horizon.

Where are we? This is the question that decides between the two geometries. In Geometry A the rebound happened everywhere, there is no centre to be far from, and the test must come up empty. In Geometry B there is a centre, and its signature is Geometry B's own prediction, P4 (Off-Centre

Flow): an off-centre dipole $\boldsymbol{\xi}$ in the structure-subtracted galaxy velocity field above the mock noise floor — the 95th percentile of the false centres that isotropic mocks manufacture on the same sky mask, about 2.5 Mpc at present — or a bulk flow that grows rather than fades with depth. We looked, and after the mock-mask and 2M++ controls the answer so far is: no offset claimed, two megaparsecs at most, with a 1.7 Mpc residual that lies inside the mock distribution and points along the CMB dipole, which one re-run — the 2M++ external dipole left free — will either confirm as calibration-independent or dissolve as a calibration. What we do have is a bulk flow of 428 ± 36 km s^{-1} at $200\,h^{-1}$ Mpc, which the kinetic Sunyaev–Zel'dovich data do not forbid at that depth but would forbid if $\gtrsim 250$ km s^{-1} of it persisted to a gigaparsec, and a sky whose extreme spots all lie in the southern hemisphere. The reading we adopt is that we are not visibly off-centre, that the lopsidedness of the sky is more plausibly the work of a neighbour than of our own position, and that Geometry B has no bounce mechanism yet, so the choice between A and B rests on the measurements of Table 2 rather than on the theory. Where we are remains the least answered of the three questions, and the one whose answer would settle the most.

Where is the second Big Bang? Follow the coldest patch of the oldest light: galactic longitude 203ř, latitude −56ř, the Cold Spot. If it is the fingerprint of a neighbouring Big Pool Bang written into the initial conditions of our last-scattering surface, then that neighbour sits just beyond it, about one billion light-years past the edge of everything we can see, with a mass of order 10^{48} kg, uncertain by a factor of four upward and two and a half downward — a dwarf beside the 2×10^{52} kg in each of our own Hubble volumes. It lies beyond our horizon, so nothing it has done since can reach us: no black holes, no gravitational-wave echoes, and an induced drift of our neighbourhood of a tenth of a kilometre per second, too small to move anything we can measure. Seven other extreme spots lie on the same side of the sky, a candidate pool of neighbours clustered rather than surrounding us — though whether a neighbour can lopside the sky's power, rather than merely share its hemisphere, is not something we have computed. The neighbour reading makes promises it can be held to. It must leave no curl-mode fingerprint at the spot, where a vortex would — P8 (Curl Test), which decides between the two readings because the Spot cannot be both. Its $1/r$ potential must leak a few microkelvin across the whole sky in a pattern fixed by the spots — P9 (Low-ℓ Tail), to be judged against a thousand Gaussian skies rather than by eye. And it must contribute nothing to the bulk flow — P10 (Induced-Velocity Bound), which it does not. The neighbour hypothesis will not survive as a story; it will survive, or fall, as the set of numbers in Table 2.

What surprised us

Five things we did not expect when we racked the balls, and one thing we were glad to lose.

A one-sided sky. All eight of the most extreme spots in the Planck map lie in the southern galactic hemisphere, none in the north, and the Cold Spot is one of them.

A spiral in the Cold Spot. The Cold Spot has the most coherent spiral winding of any of the 384 sky positions we scanned, a slow 40ř twist from centre to rim: what a frozen vortex would look like, and what a Gaussian fluctuation has no reason to do.

Neighbours push, but not outward on average. Surrounding bangs do pull on our galaxies: harder along the lines toward them, and inward in the gaps between them. Gauss's law makes the sky-averaged pull exactly zero at every radius, and the directional part is about 10^{-7} of the Hubble rate. The only thing that can push isotropically is mass inside our own volume, and the kinetic Sunyaev–Zel'dovich gate caps what our own ejecta can contribute at about 1% of the observed acceleration. That is why the paper had to hand the acceleration to a constant tension of space.

Equal-mass neighbours must be very far away. If every bang had our own mass, a neighbour at the Cold Spot distance would burn a 4 K hole in the microwave sky instead of a 168 μK dimple. So either the Cold Spot source is a bang some 10^5 times lighter than ours or it is something else entirely, and equal-mass bangs must keep a spacing of at least a thousand billion light-years — several thousand if the CMB vorticity bound applies to a tidally induced spin — so that any global spin they induce is capped near 10^{-8} of the Hubble rate, and a larger one would rule the equal-mass lattice out.

The neighbour is beyond the horizon. If it exists, nothing it has done since the bounce has reached us. Everything we can test is a fossil potential laid down before the first light.

What we got rid of: the dark-matter particle

The largest simplification the framework buys is the one it was built for. Dark matter here is not a new particle. It is the $88\% \pm 4\%$ of the original Planck-mass relics that the rebound threw outward, ordinary gravitating mass with no charge, no light, and no reason to appear in any detector but a gravitational one. Forty years of direct searches, from xenon tanks to axion cavities, have found nothing, and in this picture that is the expected result rather than an embarrassment: P5 (No Particle DM) turns the null into a prediction, and a confirmed particle detection would falsify H1 (Bounce) outright. What the framework does *not* get rid of is dark energy. The constant tension of space is adopted, its value is not derived, and Section 5.2 says so. One of the two dark sectors is explained by mechanics; the other is inherited. That is the honest scorecard.

When we will know more

Most of the decisive tests are archival and cheap. The curl search (P8 (Curl Test)) at calibrated sensitivity and the low-multipole template (P9 (Low-ℓ Tail)) can be run on Planck PR4 products within a year; a null on the first retires the vortex reading of the Cold Spot, a null on the second retires the population reading of the extremes. The JADES handedness control field is a single reprocessing of public JWST data. Our own position (P4 (Off-Centre Flow)) waits on the next CosmicFlows release and on DESI and Euclid peculiar velocities, roughly 2027–2029, which should tighten the mock floor by $\sim\sqrt{2}$ and settle whether the 1.7 Mpc residual is calibration. The polarization ring and cyclonic B-modes, P1 (Polarization Ring) and P2 (Cyclonic B-modes), need degree-scale polarization at the microkelvin-arcminute level: Simons Observatory results toward 2028, CMB-S4 and LiteBIRD in the early 2030s. The gravitational-wave background from a first-order transition, P3 (Transition GW Background), is a LISA-era question, after 2035. By the end of this decade, then, the Cold Spot will be a vortex, a neighbour, or neither, and we will know whether we sit visibly off-centre in our own Big Pool Bang.

In one paragraph: a jammed crystal that bounced, a speed-sorted debris field we live inside — read either as the seed of the Hubble flow or as the Hubble flow itself — and a cold fingerprint on the far wall that may belong to someone else. Each of these is stated so that a measurement can remove it.

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Appendix A. Exploratory Null Results and a Pre-registered Test

This appendix records the exploratory battery run on the eight most extreme large-scale spots of the Planck SMICA map ($N_{\rm side} = 128$, $|b| > 25$ř, 2ř smoothing; per-spot amplitude ΔT and half-maximum radius θ; positions in Sec. 9). Nulls are 60–300 Gaussian realizations drawn from the map’s own pseudo-C_ℓ and passed through the identical pipeline unless stated otherwise.

Locus test. If neighbours shared a common distance shell and a common formation density, their spots would follow a locus $|\Delta T| \propto \theta^3$, whereas Gaussian peaks obey BBKS statistics (Bardeen et al. 1986) with amplitude and size nearly independent. A log–log fit of $|\Delta T|$ against θ gives slope -0.00, $r = -0.01$, 27th percentile of the Gaussian null ($+0.07 \pm 0.10$). No amplitude–size correlation is detected, but the test is powerless as implemented (radii quantized to 2.5ř–5.5ř, amplitude spread $\sim 10\%$). It is reported as unresolved pending injection-recovery validation, not as an exclusion. Neighbours at a range of distances would populate a diffuse cloud rather than a locus, and one genuine neighbour among seven Gaussian impostors would not move the fit.

Internal dipoles and spin budget. Monopole-plus-dipole fits within 6ř in the local tangent frame: the eight extremes split 4/4 hot/cold with rank-matched amplitudes summing to $-11\,\mu$K against a mean $|\Delta T| \approx 162\,\mu$K (7% of Gaussian skies as balanced); opposite-sign spots lie 4.0ř closer than same-sign (null $+5.2$ř $\pm\, 16.3$ř, $p = 0.30$); nearest-neighbour spacing is no more regular than Gaussian. Converting dipoles to spin-axis proxies ($\mathbf{L} \propto \hat{n} \times \mathbf{d}$), the Cold Spot’s internal dipole ($16.4\,\mu$K/deg) is the largest by a factor 2.3 and the others weakly co-align. These fits are dominated by $\ell \sim 2$–20 gradients that Gaussian peaks generically carry and that are correlated across co-hemispheric spots, so they are upper bounds on rotation, not spin measurements; a gradient-subtracted re-fit is required before any spin fraction can be quoted.

Spiral winding. The azimuthal phase of the $m = 1$ mode in 1.5ř rings from 1.5ř to 15ř around the Cold Spot is locked ring-to-ring with coherence 0.988 — above all 300 random positions (null

0.72 ± 0.18) — and drifts by ~ 3ř of phase per degree of radius (~ 40ř total twist). A static background gradient gives high coherence but zero winding; only the winding is spin-specific. The $m = 2, 3$ modes are unremarkable, and this is a post-hoc test.

Curl search. The public PR4 lensing release (Carron, Mirmelstein & Lewis 2022) contains only gradient modes, so we implemented an approximate temperature-only quadratic estimator ($N_{\rm side} = 256$, $\ell \leq 512$, spin-1 curl projection, mean field from 40 Gaussian simulations; unnormalized, valid only as a relative comparison). Stacking curl-potential variance in 6ř discs: the eight spots sit at the 82nd percentile of 60 simulated skies and the Cold Spot at the **38th** — indistinguishable from a random position. This does not trigger the retirement clause of P8 (Curl Test), which is bound to a sensitivity reaching the temperature-dipole-implied rotation that this reduced-resolution estimator does not attain (Planck's full-depth reconstruction uses $\ell \leq 2048$ with polarization).

Polarization. On the SMICA Q/U maps ($N_{\rm side} = 256$, $\ell \leq 512$, E/B decomposition, 1ř smoothing): stacked B-mode variance in 6ř discs at the eight spots is at the 66th percentile of random positions, E-mode variance at the 35th, and the Cold Spot's radial-E profile is consistent with zero in every 2ř ring to 20ř with per-ring errors of 0.6–2.7 μK — the predicted 1–3 μK ring of P1 (Polarization Ring) is neither detected nor excluded.

Lensing convergence. The sign-weighted PR4 κ stack (Planck Collaboration 2020 VIII; Carron et al. 2022) on the eight positions is $+0.006$ against a random-position null of 0.000 ± 0.013 (4/8 sign matches): no integrated mass scars. The Cold Spot's inner-2ř $\kappa = -0.070$ vs null $+0.013 \pm 0.042$ (-2σ) has the sign of its known shallow foreground supervoid; the 8ř–20ř annulus never exceeds $+0.6\sigma$, so there is no mass "belt". A stack on $\sim$27,000 ordinary small spots (1ř smoothing, $|T| > 1.5\sigma$) shows cold spots on κ deficits (-7.2σ) and hot on excesses ($+2.2\sigma$), but κ is reconstructed from the same temperature data and quadratic estimators leak temperature extrema into the mass map; whether this is leakage or genuine ISW×lensing is undiscriminated, and a polarization-only κ stack is the required discriminating check.

CMB-independent tracers. Against the Quaia Gaia–unWISE quasar catalogue (Storey-Fisher et al. 2024; 756k quasars, $z \sim 1$–3, selection-function corrected) the small-spot correlation vanishes (cold -1.1σ, hot -0.3σ) and the eight extremes are within $\sim 1\sigma$ of random (Cold Spot -0.018, 0.6σ); this is not a sensitive null, since Quaia's kernel barely overlaps the ISW kernel. Against WISE×SuperCOSMOS (Bilicki et al. 2016; 17.7M galaxies, $z = 0.1$–0.4, high-passed at 10ř): small spots remain null (cold -0.7σ, hot $+0.2\sigma$), while the extremes show seven of eight sign matches and a sign-weighted stack of $\approx +2.7\sigma$ against a correlated-noise null from 300 random 5ř discs. Qualifications: the result rests on three sightlines ($|z| \approx 2.0$–2.3); the nominal binomial 3.5% ignores the three-tracer, multi-statistic search (global $p \sim 10$–30%); extremeness-conditioning inflates the recovered correlation (Eddington-type bias); the 10ř high-pass self-subtracts genuine $\gtrsim 10$ř supervoids; and extinction/stellar residuals at $|b| = 25$–40ř are of order the per-spot signals. Verdict: a 1–2σ-class hint that the extremes are partly sourced by $z \sim 0.1$–0.4 superstructures, to be confirmed or retired by a pre-registered re-run at two baseline scales with $|b| > 30$ř and extinction regression. The 2M++ cross-match of the eight sightlines ($< 180\,h^{-1}$ Mpc) matches spot signs at coin-flip rates (5 of 8).

JWST handedness. The JADES field at (224ř, -54ř), 8.6ř from the Cold Spot, shows a claimed 2 : 1 handedness asymmetry in 263 high-redshift galaxies (Shamir 2025). The sample is small, the method contested, and a Doppler boost from the Milky Way's rotation is a mundane candidate. The discriminating measurement is a control field — the identical pipeline on COSMOS or GOODS-N (> 100ř away): a systematic reproduces the asymmetry everywhere; a Cold Spot spin column produces it only on the column. The local anchor is already measured: the same-handedness angular correlation over 77{,}840 SDSS galaxies (11.2M pairs, 0.05ř–4ř) is zero in every bin, with a 1.9σ global excess.

Coherence-gradient hypothesis (untested). In the N-body toy, only the earliest, fastest ejecta escape in a single pass and preserve geometric correlations; everything slower undergoes violent relaxation, which conserves spin magnitude but randomizes spin geometry. If any residual radial ordering survived in our Hubble volume — H5 (Our Position) — one would expect a *coherence gradient*: fossil geometry intact in the earliest-ejected material, degrading toward the well-mixed later ejecta. Intermediate tracers (quasar-polarization alignments at $z \sim 1$–2; galaxy handedness axes) lie 24ř–73ř from the Cold Spot direction, consistent with random axes, and the local same-handedness correlation above is zero to a few parts in 10^3, as mixing requires. This is recorded as a hypothesis; it has no validated observable and is not part of the main-text claims.

Statistical accounting. Across the battery (locus slope, hemisphere count, hot/cold balance, pairing distance, lattice regularity, dipole magnitudes, cancellation ratio, winding coherence, κ stack, curl percentile, tracer stacks) the individual results are $p \sim 0.06$–0.7; under any global look-elsewhere correction the sub-2σ items are jointly consistent with noise. Further method limits: the Gaussian nulls use the unmasked pseudo-C_ℓ without mask deconvolution, anisotropic noise, or foreground residuals (end-to-end NPIPE simulations are the required upgrade for any sub-percent claim), and the spots were selected on the same map on which all statistics were evaluated, so the hypotheses were partly formulated post hoc. What survives is qualitative: the extreme-spot sky is one-sided at literature-level significance, the Cold Spot is the outlier in every channel examined, and nothing requires a population of additional objects.

Pre-registered commitment (P8 (Curl Test)). The load-bearing, falsifiable claim of the vortex reading is the curl test. We commit in advance that a clean stacked curl-mode null at the Cold Spot position, at a sensitivity reaching the temperature-dipole-implied rotation, *retires* the vortex interpretation of that direction in favor of the non-rotating neighbour reading (H3 (Neighbour)/H6 (Neighbour Location)) — not merely disfavors it. If curl variance is instead detected localized there, the vortex reading stands and the Cold Spot becomes a relic vortex of our own bang rather than a neighbour's imprint.

Appendix B. A Fixed Grid of Surrounding Bangs (Consistency Check)

The population model in its most symmetric form — the limiting case of the one-sided population H7 (Pool Model) adopts (Sec. 9e) — places Big Pool Bangs on a fixed lattice with our own at one site. As a consistency check, we ask whether the neighbours' pull on our ejecta could contribute to the observed late-time acceleration. Point masses of equal mass were placed on a cubic grid of unit spacing (26 neighbours for $3 \times 3 \times 3$, 124 for $5 \times 5 \times 5$); test galaxies were placed at distance r from us in random directions and the vector sum of the neighbours' inverse-square pulls was evaluated (Table 6). Units: the pull of one neighbour on us.

Table 6. Net pull of a cubic lattice of equal-mass neighbours on test galaxies at distance r from us.

r (grid units)	Sky-averaged outward pull	Toward a neighbour (axis)	Toward a gap (diagonal)
0.1	0.0000	+0.012	−0.008
0.2	0.000	+0.10	−0.06

r (grid units)	Sky-averaged outward pull	Toward a neighbour (axis)	Toward a gap (diagonal)
0.3	0.000	+0.34	−0.20
0.4	0.00	+0.85	−0.44
0.45	0.00	+1.26	−0.60

The results (Table 6) are identical for both grid sizes. Galaxies lying toward a neighbour are pulled outward with a pull that grows steeply with r; galaxies lying toward the gaps between neighbours are pulled inward; the sky average is zero at every radius. This is Gauss's law, not a property of the arrangement: the sky-averaged radial acceleration at radius r is $GM_{\rm enc}(r)/r^2$, and external sources enclose nothing. Only mass, or negative pressure, *inside* the observed volume can change the isotropic expansion rate. The lattice therefore predicts (i) zero contribution to the monopole acceleration, and (ii) a direction-dependent expansion rate with the cubic-harmonic ($\ell = 4$) pattern of the lattice, fast toward the six nearest neighbours and slow toward the eight diagonal gaps. Observation (ii) is a further falsifier for the lattice limit of H7 (Table 2): supernova and BAO expansion-rate anisotropy is limited to the few-percent level with no cubic pattern, while the amplitude implied by the neighbour mass and distance of Sec. 8 is $\sim 10^{-7}$ at the $r \approx 0.3$ reached by the supernova sample. The one form in which surrounding mass does raise the local expansion rate isotropically is a local underdensity, where the effect comes from the mass *missing inside*. A $\delta \approx -0.3$ underdensity within $\sim$300 Mpc, as claimed by Keenan, Barger & Cowie (2013), would imply an outflow of $\sim$1000 km s^{-1} at its edge, in tension with the kSZ gate of Sec. 5; the literature is divided (Keenan–Barger–Cowie 2013 for; Wu & Huterer 2017 and Kenworthy, Scolnic & Riess 2019 against), and this paper takes no position on whether the void exists. What it does conclude is that such a void cannot be the acceleration: a Gpc-scale void deep enough to mimic Λ is excluded by the kSZ gate. This appendix is a consistency check on the adopted expansion history of Sec. 5.2, not its basis.

Appendix C. Kinematic Mimicry of Acceleration: Tests

This appendix records the tests summarized in Sec. 5 of whether the ejecta distribution could mimic acceleration for an interior observer.

Relativistic (LTB) recalculation. Dust shells in the Lemaître–Tolman–Bondi metric obey the same radial equation as Newtonian shells; what the toy of Sec. 5.1 gets wrong is the *observables*. Integrating radial null geodesics through the evolving LTB metric (300 shells seeded from the ejecta speed distribution, fallback mass virialized into the interior, $d_L = (1+z)^2 R$) gives $q_0^{\rm eff} = +0.16$ ($w_{\rm eff}(0) \approx -0.23$): mild deceleration, no apparent acceleration, with a distance-modulus shape $\sim$0.35 mag from both ΛCDM and coasting out to $z \sim 5$. Scope restrictions: the lightcone threads only the free-streaming 9%, the interior treatment and observer epoch are modelling choices, and shell crossing was not monitored.

Profile inversion. Since the ejection profile is an input, we searched power-law, bimodal fast-tail, and stochastically lumpy profiles for any that accelerates. Smooth families robustly decelerate ($q_0^{\rm eff} \sim +1$ to $+4$); apparently negative values in lumpy configurations are seed-ensemble noise (± 1.4, exceeding the target $|q_0| \approx 0.55$, so that family is unconstrained rather than excluded). The known way an LTB model mimics acceleration — the giant void, observer inside a Gpc-scale underdensity — is the *opposite* density ordering from outward ejection.

Blast wave and the kSZ gate. A rebound acting as a blast wave would sweep ejecta into a shell and leave a rarefied cavity, the giant-void geometry; swept-shell profiles do give smoother lightcones

(0.03–0.08 mag residuals vs $\sim$1 mag). This is not a rescue: the comparison metric cannot separate ΛCDM from coasting (~ 0.04 mag rms apart), a 0.08 mag residual corresponds to $\Delta\chi^2 \sim 400$ against Pantheon+, a cavity deep enough for $q_0 \approx -0.55$ is the giant-void model excluded by kinetic Sunyaev–Zel'dovich and Compton-y constraints (Zhang & Stebbins 2011), and mass swept into a distant shell is unavailable as local dark matter. Imposing the constraints first: the kSZ limit on coherent cluster flows ($v \lesssim 300$ km s^{-1} at Gpc distances) caps the mean interior contrast of any cavity at $|\bar{\delta}| \lesssim 0.05$ (500 Mpc) to $\lesssim 0.01$ (3 Gpc), i.e. $|\Delta q_0| \lesssim 0.01$ against the $\Delta q_0 \approx -1.05$ required. No ejection profile of any morphology supplies more than $\sim$1% of the observed dark energy; this is the result quoted in Sec. 5.

A framework-native candidate, with its cost. If crystallization is incomplete today, a residual unconverted vacuum fraction would be smooth, non-clumping, and immune to the kSZ gate — qualitatively dark energy. Quantitatively this re-imports the cosmological-constant problem verbatim ($\rho_P \sim 10^{96}$ vs 6×10^{-27} kg m^{-3}: tuning to one part in 10^{123}), and ongoing conversion would imply present-epoch nucleation, domain walls (the Zel'dovich, Kobzarev & Okun 1974 bound), and an unconverted component at BBN and recombination — none computed here. The boiling picture does anchor H3 (Neighbour)–H4 (Population) in Coleman & De Luccia (1980) bubble nucleation, whose collisions imprint disc-shaped CMB features with specific polarization profiles (Aguirre & Johnson 2011) and, generically, source the stochastic gravitational-wave background of P3 (Transition GW Background).